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Total Marks: 100

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DevOps and Cloud Computing

Assignment 2

**Sakila Flask Application | End-to-End
DevOps Pipeline**



Course Lecturer: Muhammad Ali

Section: A

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Assignment 2

Sakila Flask Application | End-to-End DevOps Pipeline

Introduction

This report documents the DevOps workflow completed for the Sakila Flask application. The assignment was divided into three main parts: Git and GitHub workflow management, Docker containerization and orchestration, and GitHub Actions CI/CD automation.

The goal was to demonstrate a complete software delivery pipeline from branch-based development and pull requests through containerized deployment and automated testing.

Part A: Git Workflow and Branch Protection

Part A focused on establishing a professional Git workflow. This included forking the upstream repository, setting up remote configuration, creating feature branches, enforcing branch protection, resolving merge conflicts, conducting pull request reviews, and tagging releases.

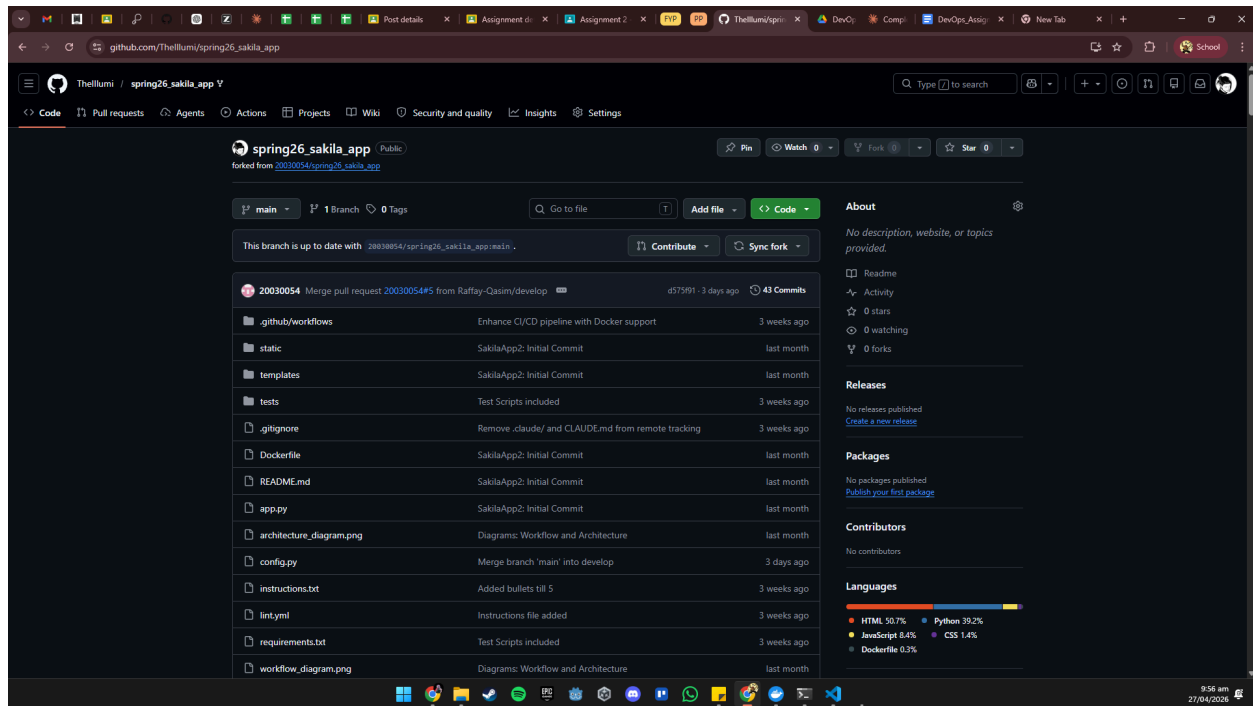
A1 | Repository Setup and Branching Strategy

The upstream repository was forked to create a personal copy. The fork was cloned locally and the remote was verified to point to the correct GitHub URL. Four branches were created and pushed to the remote:

- develop - main integration branch
- feature/docker-optimization - Docker improvements

- feature/ci-pipeline - CI/CD workflows
- hotfix/bugfix - hotfix branch

Branch protection was configured on the main branch to require pull requests before any merge, preventing direct commits to main.



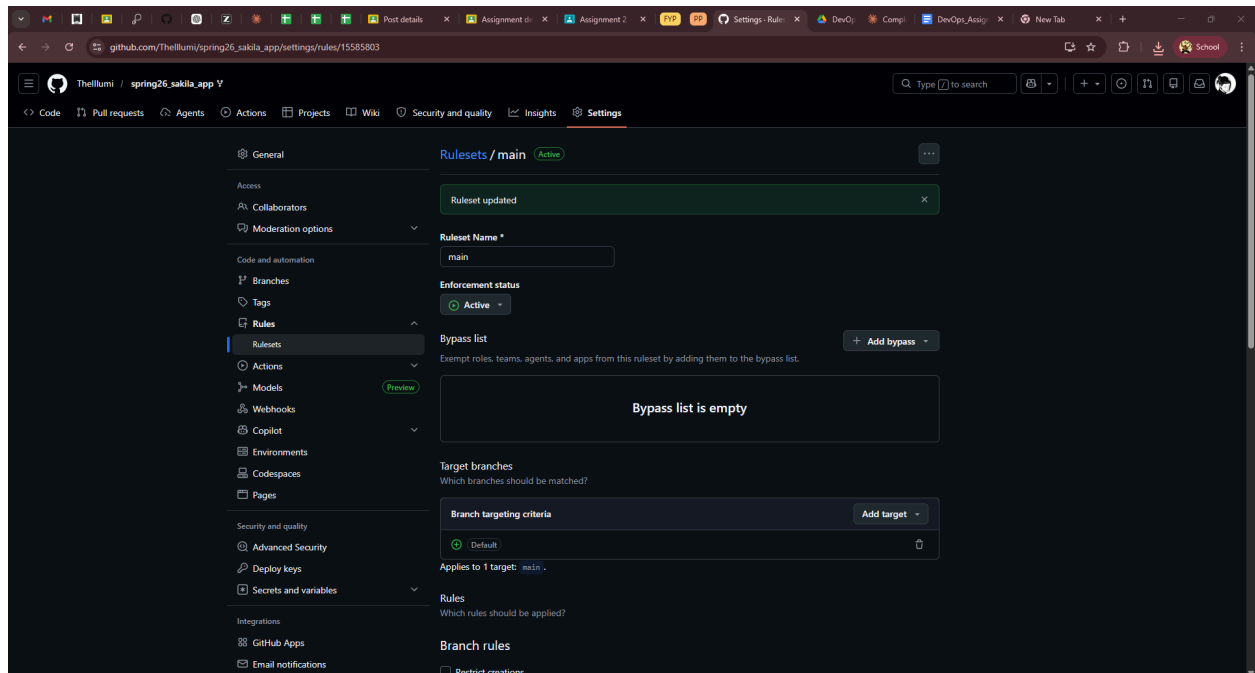
[Screenshot: A1 - Forked repository showing username in URL]

```
PowerShell
Docker Compose version v5.1.2
PS C:\Users\saadm> git config --global user.name "Saad Mughal"
PS C:\Users\saadm> git config --global user.email "saadmughal321@gmail.com"
PS C:\Users\saadm> code --version
1.116.0
560a9dba96f961efea7b1612916f89e5d5d4d679
x64
PS C:\Users\saadm> cd D:
PS D:\> cd github
PS D:\GitHub> git clone https://github.com/Saad Mughal/spring26_sakila_app.git
Cloning into 'Mughal/spring26_sakila_app.git'...
remote: Not Found
fatal: repository 'https://github.com/Saad/' not found
PS D:\GitHub> git clone https://github.com/SaadMughal/spring26_sakila_app.git
Cloning into 'spring26_sakila_app'...
remote: Repository not found.
fatal: repository 'https://github.com/SaadMughal/spring26_sakila_app.git/' not found
PS D:\GitHub> git clone https://github.com/TheIllumi/spring26_sakila_app.git
Cloning into 'spring26_sakila_app'...
remote: Enumerating objects: 166, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (20/20), done.
remote: Total 166 (delta 57), reused 49 (delta 49), pack-reused 97 (from 1)
Receiving objects: 100% (166/166), 1.71 MiB | 3.35 MiB/s, done.
Resolving deltas: 100% (73/73), done.
PS D:\GitHub> cd spring26_sakila_app
PS D:\GitHub\spring26_sakila_app> git remote -v
origin https://github.com/TheIllumi/spring26_sakila_app.git (fetch)
origin https://github.com/TheIllumi/spring26_sakila_app.git (push)
PS D:\GitHub\spring26_sakila_app> |
```

[Screenshot: A1 - git remote -v output confirming fork URL]

```
PowerShell
* [new branch]      develop -> develop
branch 'develop' set up to track 'origin/develop'.
PS D:\GitHub\spring26_sakila_app> git checkout -b feature/docker-optimization
Switched to a new branch 'feature/docker-optimization'
PS D:\GitHub\spring26_sakila_app> git push -u origin feature/docker-optimization
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/docker-optimization' on GitHub by visiting:
remote:      https://github.com/TheIllumi/spring26_sakila_app/pull/new/feature/docker-optimization
remote:
To https://github.com/TheIllumi/spring26_sakila_app.git
* [new branch]      feature/docker-optimization -> feature/docker-optimization
branch 'feature/docker-optimization' set up to track 'origin/feature/docker-optimization'.
PS D:\GitHub\spring26_sakila_app> git checkout -b feature/ci-pipeline
Switched to a new branch 'feature/ci-pipeline'
PS D:\GitHub\spring26_sakila_app> git push -u origin feature/ci-pipeline
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/ci-pipeline' on GitHub by visiting:
remote:      https://github.com/TheIllumi/spring26_sakila_app/pull/new/feature/ci-pipeline
remote:
To https://github.com/TheIllumi/spring26_sakila_app.git
* [new branch]      feature/ci-pipeline -> feature/ci-pipeline
branch 'feature/ci-pipeline' set up to track 'origin/feature/ci-pipeline'.
PS D:\GitHub\spring26_sakila_app> git checkout -b hotfix/bugfix
Switched to a new branch 'hotfix/bugfix'
PS D:\GitHub\spring26_sakila_app> git push -u origin hotfix/bugfix
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'hotfix/bugfix' on GitHub by visiting:
remote:      https://github.com/TheIllumi/spring26_sakila_app/pull/new/hotfix/bugfix
remote:
To https://github.com/TheIllumi/spring26_sakila_app.git
* [new branch]      hotfix/bugfix -> hotfix/bugfix
branch 'hotfix/bugfix' set up to track 'origin/hotfix/bugfix'.
PS D:\GitHub\spring26_sakila_app> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\GitHub\spring26_sakila_app> git branch -a
  develop
  feature/ci-pipeline
  feature/docker-optimization
  hotfix/bugfix
* main
remotes/origin/HEAD -> origin/main
remotes/origin/develop
remotes/origin/feature/ci-pipeline
remotes/origin/feature/docker-optimization
remotes/origin/hotfix/bugfix
remotes/origin/main
PS D:\GitHub\spring26_sakila_app>
```

[Screenshot: A1 - git branch -a listing all branches]



[Screenshot: A1 – GitHub branch protection settings for main]

A2 | Merge Conflict Creation and Resolution

Two feature branches, feature/update-config and feature/add-healthcheck, were created from develop. Both branches modified the same section of config.py, introducing a conflict when the second branch was merged.

The conflict occurred because one branch set MYSQL_HOST to 'sakila-db-server' while the other set it to 'db-primary'. The conflict was resolved manually by keeping 'sakila-db-server' as the host and preserving both CONNECTION_TIMEOUT and HEALTH_CHECK_INTERVAL variables in the final file.

```
1 # <<<<<< HEAD (Current Change)
2 <<<<<< HEAD (Current Change)
3 # Developer: Saad Mughal
4 # Date: 27-04-2020
5 # <<<<<< HEAD (Current Change)
6
7 # Name: Abdul Raffay Qasim
8 # Name: Team Member
9 # Date: 2020-04-23
10 # Minor improvement after PR review
11 # Added after review feedback
12
13
14 # Team Member: TEAMMATE NAME
15 # Date: TODAY'S DATE
16 # <<<<<< HEAD (Current Change)
17
18 >>>>>> feature/add-healthcheck (Incoming Change)
19 import os
20
21 You 59 seconds ago [2 authors (You and one other)]
22 class Config:
23     """Base configuration class for Sakila Flask application."""
24     MYSQL_HOST = os.environ.get('MYSQL_HOST', 'db-primary')
25     MYSQL_USER = os.environ.get('MYSQL_USER', 'root')
26     MYSQL_PASSWORD = os.environ.get('MYSQL_PASSWORD', 'admin')
27     MYSQL_DB = os.environ.get('MYSQL_DB', 'sakila')
28
29     try:
30         CONNECTION_TIMEOUT = int(os.environ.get('CONNECTION_TIMEOUT', '30'))
31     except ValueError:
32         CONNECTION_TIMEOUT = 30
33
34 <<<<<< HEAD (Current Change)
35 SECRET_KEY = os.environ.get('SECRET_KEY', 'your-secret-key-here-change-this-in-production')
36
37 # Author: Umrah Ta
38 # Date: 2020-04-23
39 # Purpose: Database configuration for Sakila Flask Application
40
41 # Author: Team Member = Aliyah Cheema
42 # Date: 2020-04-23
43 # Purpose: Health check configuration merged from feature/add-healthcheck
44
45 MYSQL_HOST = os.environ.get('MYSQL_HOST', 'sakila-db-server')
```

[Screenshot: A2 - config.py showing raw conflict markers (<<<<<< HEAD / ===== / >>>>>>)]

```
1 # <<<<<< HEAD (Current Change)
2 # Developer: Saad Mughal - Date: 27-04-2020
3 # Conflict resolved: kept sakila-db-server as host,
4 # Included both CONNECTION_TIMEOUT and HEALTH_CHECK_INTERVAL
5 # <<<<<< HEAD (Current Change)
6
7 import os
8
9 You 3 minutes ago [4 authors (You and others)]
10 class Config:
11     """Base configuration class for Sakila Flask application."""
12     MYSQL_HOST = os.environ.get('MYSQL_HOST', 'sakila-db-server')
13     MYSQL_USER = os.environ.get('MYSQL_USER', 'root')
14     MYSQL_PASSWORD = os.environ.get('MYSQL_PASSWORD', 'admin')
15     MYSQL_DB = os.environ.get('MYSQL_DB', 'sakila')
16
17     try:
18         CONNECTION_TIMEOUT = int(os.environ.get('CONNECTION_TIMEOUT', '30'))
19     except ValueError:
20         CONNECTION_TIMEOUT = 30
21
22     try:
23         HEALTH_CHECK_INTERVAL = int(os.environ.get('HEALTH_CHECK_INTERVAL', '10'))
24     except ValueError:
25         HEALTH_CHECK_INTERVAL = 10
26
27     SECRET_KEY = os.environ.get(
28         'SECRET_KEY',
29         'your-secret-key-here-change-this-in-production'
30     )
31
32 You 3 minutes ago [Uncommitted changes]
```

[Screenshot: A2 - Resolved config.py with both variables and combined comment block]


```

PowerShell
* a7b4f29 (HEAD -> develop) merge: resolve conflict between update-config and add-healthcheck Confl
ct was in config.py MYSQL_HOST default value and comment block. Resolution: kept 'sakila-db-server' as
host (from feature/update-config), combined both comment blocks, included both new variables.
| \
| * 6fdb3b1 (feature/add-healthcheck) feat(config): add health check interval and update db host
* | 6ff1051 (feature/update-config) feat(config): update database host and add timeout setting
| /
* d575f91 (origin/main, origin/hotfix/bugfix, origin/feature/docker-optimization, origin/feature/ci-
pipeline, origin/develop, origin/HEAD, main, hotfix/bugfix, feature/docker-optimization, feature/ci-pi
pipeline) Merge pull request #5 from Raffay-Qasim/develop
| \
| * 0c731c6 Merge branch 'main' into develop
| /
* c773b0d Merge pull request #4 from f2023-743-hub/develop
| \
| * 53da296 Merge branch 'main' into develop
| /
* 09b3821 Merge pull request #1 from humzamajeed/develop
| \
| * 1b4ab70 fix(config): reduce default CONNECTION_TIMEOUT to 10s per review feedback
* | 3cc14a3 merge: resolve conflict between feature/update-config and feature/add-healthcheck
| /
| * c9a37ac feat(config): update database host and add health check interval
| /
| \
| * 7dda211 added my name which i forgot earlier
* | c0e7a64 feat(config): update database host and add timeout setting
| /
| \
| * 8c28b00 fix(config): add input validation for timeout environment variables
* | 1a8dc55 fix(merge): resolve conflict between feature/update-config and feature/add-healthcheck
| /
| * 831daf7 feat(config): add health check interval and update host to db-primary
| /
| \
| * 17b35a5 feat(config):update database host and add timeout setting
| /
| \
| * 513d6bd fix: address review feedback
* | d9dc7eb chore: minor improvement after review
* | 9d53eff docs: add class docstring to address review feedback
* | 7377069 fix(config): resolve merge conflict and combine timeout/healthcheck settings
| \
| * 3492408 feat(config): add health check interval and update db host
| /
| \
| * 13d2bfa feat(config): update database host and add timeout setting
| /
:

```

```
PowerShell
PS D:\GitHub\spring26_sakila_app> cat config.py
# =====
# Developer: Saad Mugha - Date: 27-04-2026
# Conflict resolved: kept sakila-db-server as host,
# included both CONNECTION_TIMEOUT and HEALTH_CHECK_INTERVAL
# =====
import os

class Config:
    """Base configuration class for Sakila Flask application."""

    MYSQL_HOST = os.environ.get('MYSQL_HOST', 'sakila-db-server')
    MYSQL_USER = os.environ.get('MYSQL_USER', 'root')
    MYSQL_PASSWORD = os.environ.get('MYSQL_PASSWORD', 'admin')
    MYSQL_DB = os.environ.get('MYSQL_DB', 'sakila')

    try:
        CONNECTION_TIMEOUT = int(os.environ.get('CONNECTION_TIMEOUT', '30'))
    except ValueError:
        CONNECTION_TIMEOUT = 30

    try:
        HEALTH_CHECK_INTERVAL = int(os.environ.get('HEALTH_CHECK_INTERVAL', '10'))
    except ValueError:
        HEALTH_CHECK_INTERVAL = 10

    SECRET_KEY = os.environ.get(
        'SECRET_KEY',
        'your-secret-key-here-change-this-in-production'
    )

PS D:\GitHub\spring26_sakila_app> git log --oneline
a7b4f29 (HEAD -> develop) merge: resolve conflict between update-config and add-healthcheck Conflict w
as in config.py MYSQL_HOST default value and comment block. Resolution: kept 'sakila-db-server' as hos
t (from feature/update-config), combined both comment blocks, included both new variables.
6fdb3b1 (feature/add-healthcheck) feat(config): add health check interval and update db host
6ff1051 (feature/update-config) feat(config): update database host and add timeout setting
d575f91 (origin/main, origin/hotfix/bugfix, origin/feature/docker-optimization, origin/feature/ci-pipe
line, origin/develop, origin/HEAD, main, hotfix/bugfix, feature/docker-optimization, feature/ci-pipeli
ne) Merge pull request #5 from Raffay-Qasim/develop
0c731c6 Merge branch 'main' into develop
c773b0d Merge pull request #4 from f2023-743-hub/develop
53da296 Merge branch 'main' into develop
09b3821 Merge pull request #1 from humzamajeed/develop
513d6bd fix: address review feedback
d9dc7eb chore: minor improvement after review
8c28b00 fix(config): add input validation for timeout environment variables
1a8dc55 fix(merge): resolve conflict between feature/update-config and feature/add-healthcheck
831daf7 feat(config): add health check interval and update host to db-primary
17b35a5 feat(config):update database host and add timeout setting
9d53eff docs: add class docstring to address review feedback
7377069 fix(config): resolve merge conflict and combine timeout/healthcheck settings
```

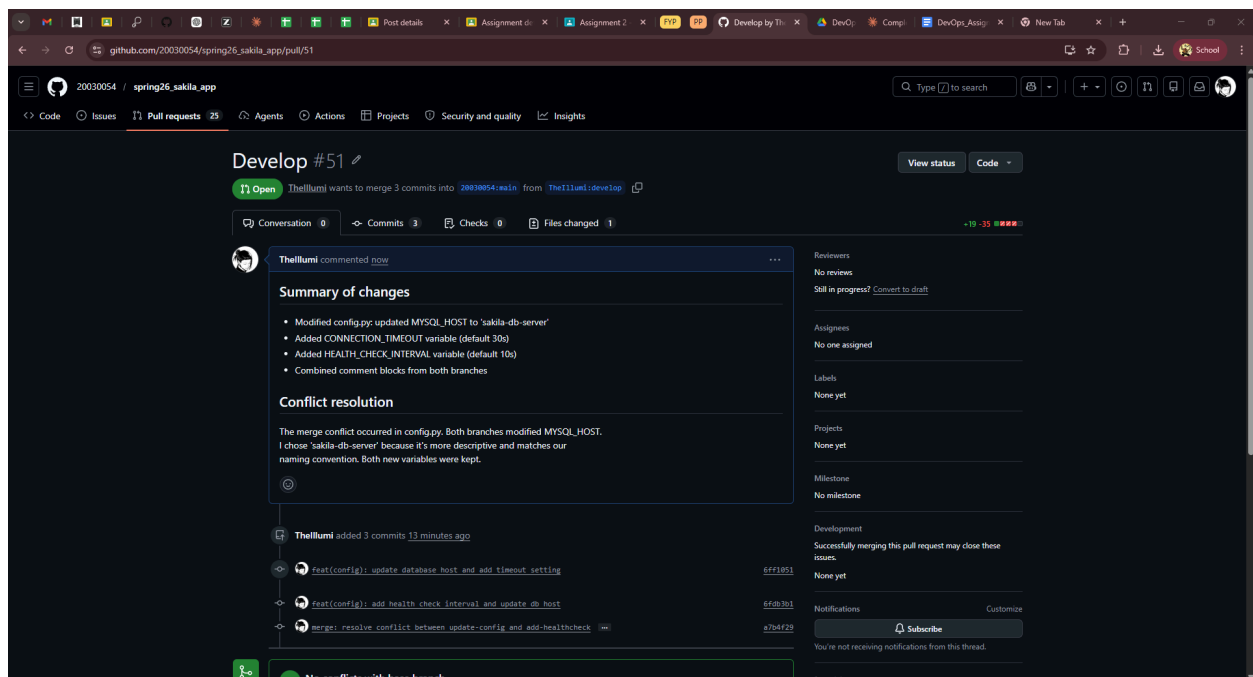
```
PowerShell
a7b4f29 (HEAD -> develop) merge: resolve conflict between update-config and add-healthcheck Conflict w
as in config.py MYSQL_HOST default value and comment block. Resolution: kept 'sakila-db-server' as hos
t (from feature/update-config), combined both comment blocks, included both new variables.
6fdb3b1 (feature/add-healthcheck) feat(config): add health check interval and update db host
6ff1051 (feature/update-config) feat(config): update database host and add timeout setting
d575f91 (origin/main, origin/hotfix/bugfix, origin/feature/docker-optimization, origin/feature/ci-pipe
line, origin/develop, origin/HEAD, main, hotfix/bugfix, feature/docker-optimization, feature/ci-pipeli
ne) Merge pull request #5 from Raffay-Qasim/develop
0c731c6 Merge branch 'main' into develop
c773b0d Merge pull request #4 from f2023-743-hub/develop
53da296 Merge branch 'main' into develop
09b3821 Merge pull request #1 from humzamajeed/develop
513d6bd fix: address review feedback
d9dc7eb chore: minor improvement after review
8c28b00 fix(config): add input validation for timeout environment variables
1a8dc55 fix(merge): resolve conflict between feature/update-config and feature/add-healthcheck
831daf7 feat(config): add health check interval and update host to db-primary
17b35a5 feat(config):update database host and add timeout setting
9d53eff docs: add class docstring to address review feedback
7377069 fix(config): resolve merge conflict and combine timeout/healthcheck settings
3492408 feat(config): add health check interval and update db host
13d2bfa feat(config): update database host and add timeout setting
1b4ab70 fix(config): reduce default CONNECTION_TIMEOUT to 10s per review feedback
3cc14a3 merge: resolve conflict between feature/update-config and feature/add-healthcheck
7dda211 added my name which i forgot earlier
c9a37ac feat(config): update database host and add health check interval
c0e7a64 feat(config): update database host and add timeout setting
1968509 Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
b418172 Added bullets till 5
f1fe2ea Enhance CI/CD pipeline with Docker support
048f52b Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
a7df40c Updated the instruction file
6835053 Refactor GitHub Actions workflow for Python project
3ce8531 Update main.yml
2963a6b Update CI pipeline to include compilation and testing
9fde00a Test Scripts included
17d162f Edit instructions file
e64149b Delete .github/workflows/.flake8
b100592 Delete .github/workflows/lint.yml
3d34a28 Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
cc66a5f Instructions file added
5877e4d Added Flask
f736638 Update Flake8 command with new options
207fbdb Create linting workflow for Python files
bd37f02 Add CI pipeline configuration in main.yml
26c404f Delete .github/workflows/pylint.yml
8d38216 Add Pylint workflow for Python code analysis
80221a7 Remove .claude/ and CLAUDE.md from remote tracking
5cac8c5 Diagrams: Workflow and Architecture
1859eb7 SakilaApp2: Initial Commit
:|
```

[Screenshot: A2 - git log --graph --oneline --all showing branch and merge history]

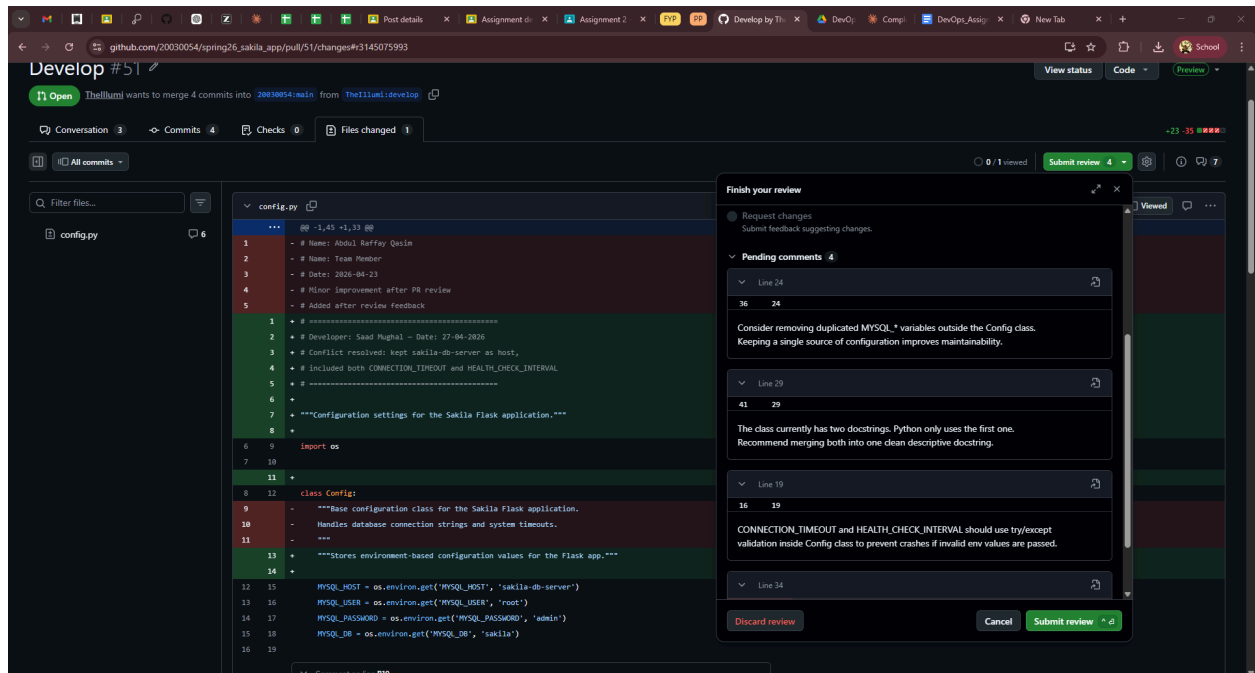
A3 | Pull Request Workflow and Code Review

A pull request was opened to merge develop into main. The PR description included a summary of all changes made, the conflict resolution approach, and a testing checklist. Inline review comments were added to key lines in config.py to explain configuration decisions.

One review comment was addressed by adding a docstring to config.py and pushing an additional commit. The PR was then merged using Squash and Merge to keep the commit history clean.



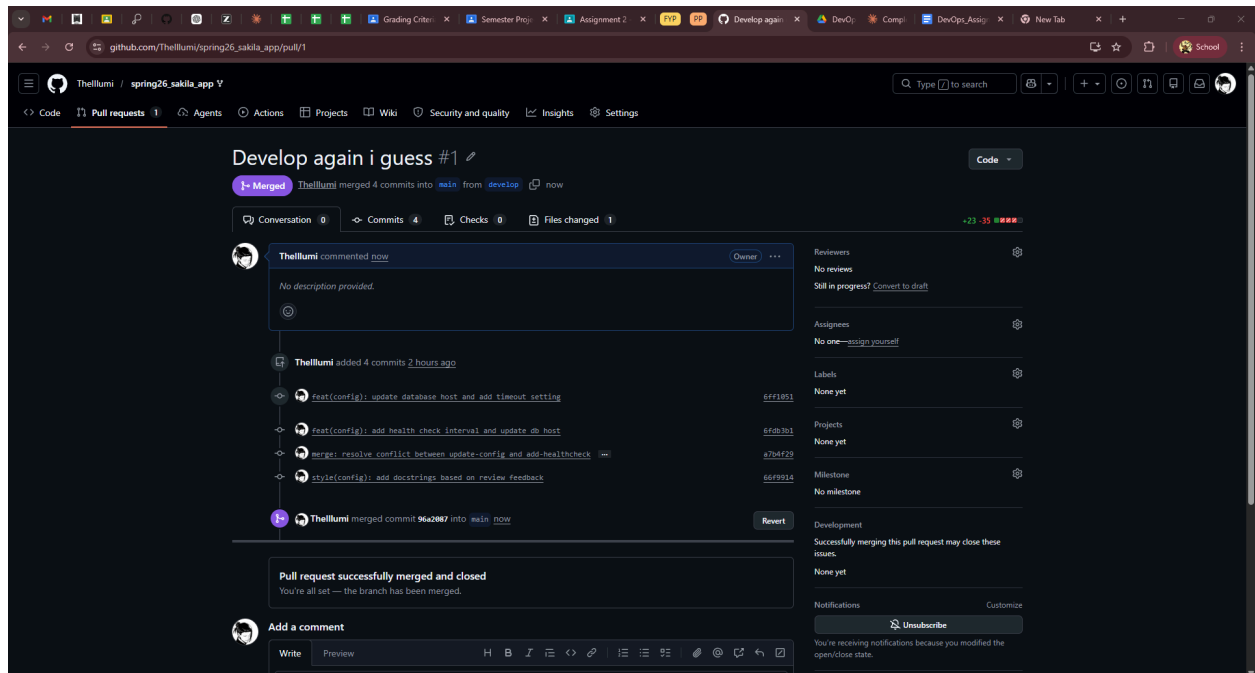
[Screenshot: A3 – PR description showing bullet list, conflict explanation, and testing checklist]



[Screenshot: A3 – Files Changed tab showing at least 3 inline review comments]

```
PowerShell
6835053 Refactor GitHub Actions workflow for Python project
3ce8531 Update main.yml
2963a6b Update CI pipeline to include compilation and testing
9fde00a Test Scripts included
17d162f Edit instructions file
e64149b Delete .github/workflows/.flake8
b100592 Delete .github/workflows/lint.yml
3d34a28 Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
cc66a5f Instructions file added
5877e4d Added Flask
f736638 Update Flake8 command with new options
207fbdb Create linting workflow for Python files
bd37f02 Add CI pipeline configuration in main.yml
26c404f Delete .github/workflows/pylint.yml
8d38216 Add Pylint workflow for Python code analysis
80221a7 Remove .claude/ and CLAUDE.md from remote tracking
5cac8c5 Diagrams: Workflow and Architecture
1859eb7 SakilaApp2: Initial Commit
PS D:\GitHub\spring26_sakila_app> git checkout develop
Already on 'develop'
Your branch is ahead of 'origin/develop' by 3 commits.
(use "git push" to publish your local commits)
PS D:\GitHub\spring26_sakila_app> git push origin develop
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.90 KiB | 1.90 MiB/s, done.
Total 9 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 2 local objects.
To https://github.com/TheIllumi/spring26_sakila_app.git
d575f91..a7b4f29 develop -> develop
PS D:\GitHub\spring26_sakila_app> git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.
PS D:\GitHub\spring26_sakila_app> git add config.py
PS D:\GitHub\spring26_sakila_app> git commit -m "style(config): add docstrings based on review feedback"
[develop 66f9914] style(config): add docstrings based on review feedback
1 file changed, 6 insertions(+), 2 deletions(-)
PS D:\GitHub\spring26_sakila_app> git push origin develop
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 556 bytes | 556.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/TheIllumi/spring26_sakila_app.git
a7b4f29..66f9914 develop -> develop
PS D:\GitHub\spring26_sakila_app> |
```

[Screenshot: A3 - New commit in PR timeline addressing a review comment]



[Screenshot: A3 – Squash merge confirmation on GitHub]

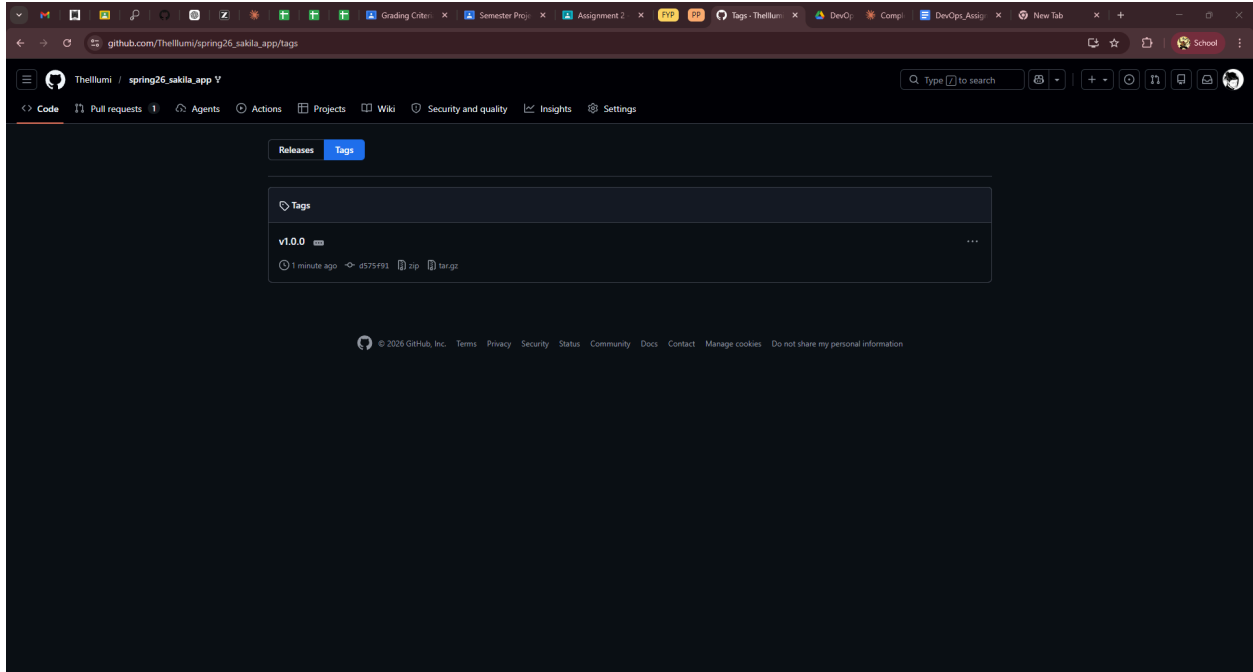
A4 | Upstream Sync and Annotated Tagging

The upstream remote was added and verified using `git remote -v`. The upstream main branch was fetched to check for any commits ahead of the fork. An annotated tag `v1.0.0` was created on main to mark the completion of Part A.

```
PowerShell
3d34a28 Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
cc66a5f Instructions file added
5877e4d Added Flask
f736638 Update Flake8 command with new options
207fbd8 Create linting workflow for Python files
bd37f02 Add CI pipeline configuration in main.yml
26c404f Delete .github/workflows/pylint.yml
8d38216 Add Pylint workflow for Python code analysis
80221a7 Remove .claude/ and CLAUDE.md from remote tracking
5cac8c5 Diagrams: Workflow and Architecture
1859eb7 SakilaApp2: Initial Commit
PS D:\GitHub\spring26_sakila_app> git checkout develop
Already on 'develop'
Your branch is ahead of 'origin/develop' by 3 commits.
(use "git push" to publish your local commits)
PS D:\GitHub\spring26_sakila_app> git push origin develop
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.90 KiB | 1.90 MiB/s, done.
Total 9 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 2 local objects.
To https://github.com/TheIllumi/spring26_sakila_app.git
d575f91..a7b4f29 develop -> develop
PS D:\GitHub\spring26_sakila_app> git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.
PS D:\GitHub\spring26_sakila_app> git add config.py
PS D:\GitHub\spring26_sakila_app> git commit -m "style(config): add docstrings based on review feedback"
[develop 66f9914] style(config): add docstrings based on review feedback
1 file changed, 6 insertions(+), 2 deletions(-)
PS D:\GitHub\spring26_sakila_app> git push origin develop
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 556 bytes | 556.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/TheIllumi/spring26_sakila_app.git
a7b4f29..66f9914 develop -> develop
PS D:\GitHub\spring26_sakila_app> git remote add upstream https://github.com/20030054/spring26_sakila_app.git
PS D:\GitHub\spring26_sakila_app> git remote -v
origin https://github.com/TheIllumi/spring26_sakila_app.git (fetch)
origin https://github.com/TheIllumi/spring26_sakila_app.git (push)
upstream https://github.com/20030054/spring26_sakila_app.git (fetch)
upstream https://github.com/20030054/spring26_sakila_app.git (push)
PS D:\GitHub\spring26_sakila_app> |
```

[Screenshot: A4 - git remote -v showing both origin (fork) and upstream (original repo)]


```
PowerShell
To https://github.com/TheIllumi/spring26_sakila_app.git
* [new tag]          v1.0.0 -> v1.0.0
PS D:\GitHub\spring26_sakila_app> git tag -l -n1
v1.0.0          Assignment Part A Complete - Git Workflow Established
PS D:\GitHub\spring26_sakila_app> git log --oneline --decorate
d575f91 (HEAD -> main, tag: v1.0.0, upstream/main, origin/main, origin/hotfix/bugfix, origin/feature/docker-optimization, origin/feature/ci-pipeline, origin/HEAD, hotfix/bugfix, feature/docker-optimization, feature/ci-pipeline) Merge pull request #5 from Raffay-Qasim/develop
0c731c6 Merge branch 'main' into develop
c773b0d Merge pull request #4 from f2023-743-hub/develop
53da296 Merge branch 'main' into develop
09b3821 Merge pull request #1 from humzamajeed/develop
513d6bd fix: address review feedback
d9dc7eb chore: minor improvement after review
8c28b00 fix(config): add input validation for timeout environment variables
1a8dc55 fix(merge): resolve conflict between feature/update-config and feature/add-healthcheck
831da7f feat(config): add health check interval and update host to db-primary
17b35a5 feat(config):update database host and add timeout setting
9d53eff docs: add class docstring to address review feedback
7377069 fix(config): resolve merge conflict and combine timeout/healthcheck settings
3492408 feat(config): add health check interval and update db host
13d2bfa feat(config): update database host and add timeout setting
1b4ab70 fix(config): reduce default CONNECTION_TIMEOUT to 10s per review feedback
3cc14a3 merge: resolve conflict between feature/update-config and feature/add-healthcheck
7dda211 added my name which i forgot earlier
c9a37ac feat(config): update database host and add health check interval
c0e7a64 feat(config): update database host and add timeout setting
1968509 Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
b418172 Added bullets till 5
f1fe2ea Enhance CI/CD pipeline with Docker support
048f52b Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
a7df40c Updated the instruction file
6835053 Refactor GitHub Actions workflow for Python project
3ce8531 Update main.yml
2963a6b Update CI pipeline to include compilation and testing
9fde00a Test Scripts included
17d162f Edit instructions file
e64149b Delete .github/workflows/.flake8
b100592 Delete .github/workflows/lint.yml
3d34a28 Merge branch 'main' of https://github.com/20030054/spring26_sakila_app
cc66a5f Instructions file added
5877e4d Added Flask
f736638 Update Flake8 command with new options
207fbdb Create linting workflow for Python files
bd37f02 Add CI pipeline configuration in main.yml
26c404f Delete .github/workflows/pylint.yml
8d38216 Add Pylint workflow for Python code analysis
80221a7 Remove .claude/ and CLAUDE.md from remote tracking
5cac8c5 Diagrams: Workflow and Architecture
1859eb7 SakilaApp2: Initial Commit
PS D:\GitHub\spring26_sakila_app> |
```



[Screenshot: A4 - git tag -l -n1 and git log --online --decorate showing the v1.0.0 tag]

Part B: Docker Containerization

Part B covered Docker containerization of the Sakila Flask application. Tasks included Dockerfile analysis and optimization, custom networking, cross-network isolation, data persistence using volumes, and full-stack orchestration using Docker Compose.

B1 | Dockerfile Analysis and Optimization

A broken Dockerfile (Dockerfile.broken) was kept for comparison. Six issues were identified in it:

- Base image too large, python:3.9 (~900 MB) replaced with python:3.9-slim (~150 MB)

- Layer caching broken, application code was copied before requirements, invalidating cache on every code change
- Multiple RUN commands, combined into a single pip install to reduce image layers
- Hardcoded credentials, MYSQL_PASSWORD was embedded directly in the image
- Unnecessary ports, ports 3306 and 22 were exposed but not needed by the Flask app
- Runs as root, no non-root user was defined; fixed by creating an appuser

The optimized Dockerfile added Docker labels, a health check, a non-root user, and proper layer ordering. The resulting image was significantly smaller than the broken version.

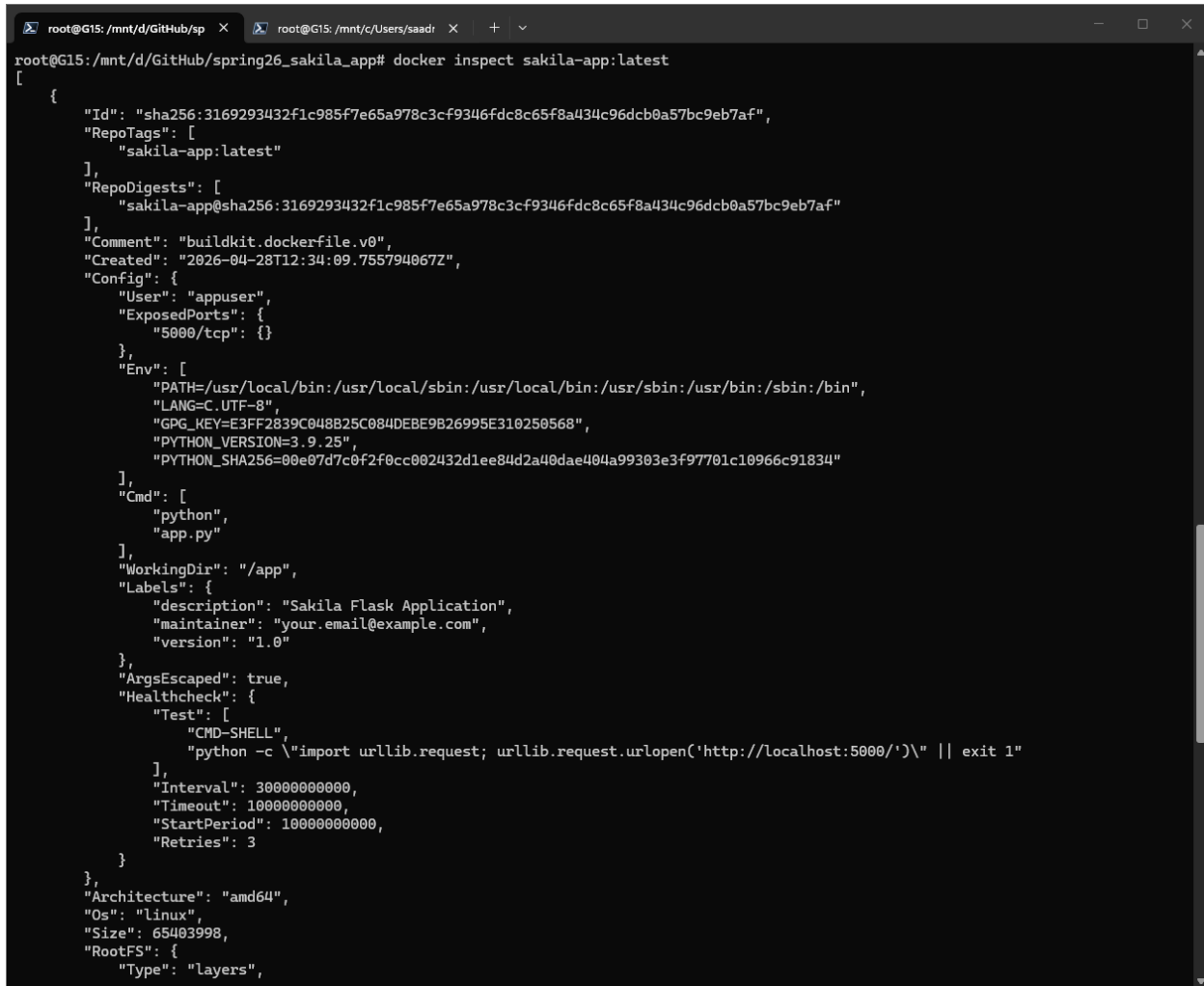
```

root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad: X + v
[General Feedback]
The most similar commands are:
  ► gres

PS D:\GitHub\spring26_sakila_app> docker images | gres sakila
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
Get-InstalledPSResource: The input object cannot be bound to any parameters for the command either because the command d
oes not take pipeline input or the input and its properties do not match any of the parameters that take pipeline input.
PS D:\GitHub\spring26_sakila_app> wsl
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker images | grep sakila
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
sakila-app:latest          3169293432f1      256MB      65.4MB
sakila-broken:latest       25114f109944      1.65GB     426MB
spring26_sakila_app-app:latest  e1572e79f688      258MB      65.6MB
spring26_sakila_app-test-runner:latest c97e54db96b0      258MB      65.6MB   U
root@G15:/mnt/d/GitHub/spring26_sakila_app#

```

[Screenshot: B1 - docker images showing both sakila-broken and sakila-app with size comparison]



```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker inspect sakila-app:latest
[
  {
    "Id": "sha256:3169293432f1c985f7e65a978c3cf9346fdc8c65f8a434c96dcb0a57bc9eb7af",
    "RepoTags": [
      "sakila-app:latest"
    ],
    "RepoDigests": [
      "sakila-app@sha256:3169293432f1c985f7e65a978c3cf9346fdc8c65f8a434c96dcb0a57bc9eb7af"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-28T12:34:09.755794067Z",
    "Config": {
      "User": "appuser",
      "ExposedPorts": {
        "5000/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
        "PYTHON_VERSION=3.9.25",
        "PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee84d2a40dae404a99303e3f97701c10966c91834"
      ],
      "Cmd": [
        "python",
        "app.py"
      ],
      "WorkingDir": "/app",
      "Labels": {
        "description": "Sakila Flask Application",
        "maintainer": "your.email@example.com",
        "version": "1.0"
      },
      "ArgsEscaped": true,
      "Healthcheck": {
        "Test": [
          "CMD-SHELL",
          "python -c \"import urllib.request; urllib.request.urlopen('http://localhost:5000/')\" || exit 1"
        ],
        "Interval": 3000000000,
        "Timeout": 1000000000,
        "StartPeriod": 1000000000,
        "Retries": 3
      }
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 65403998,
    "RootFS": {
      "Type": "Layers",
```

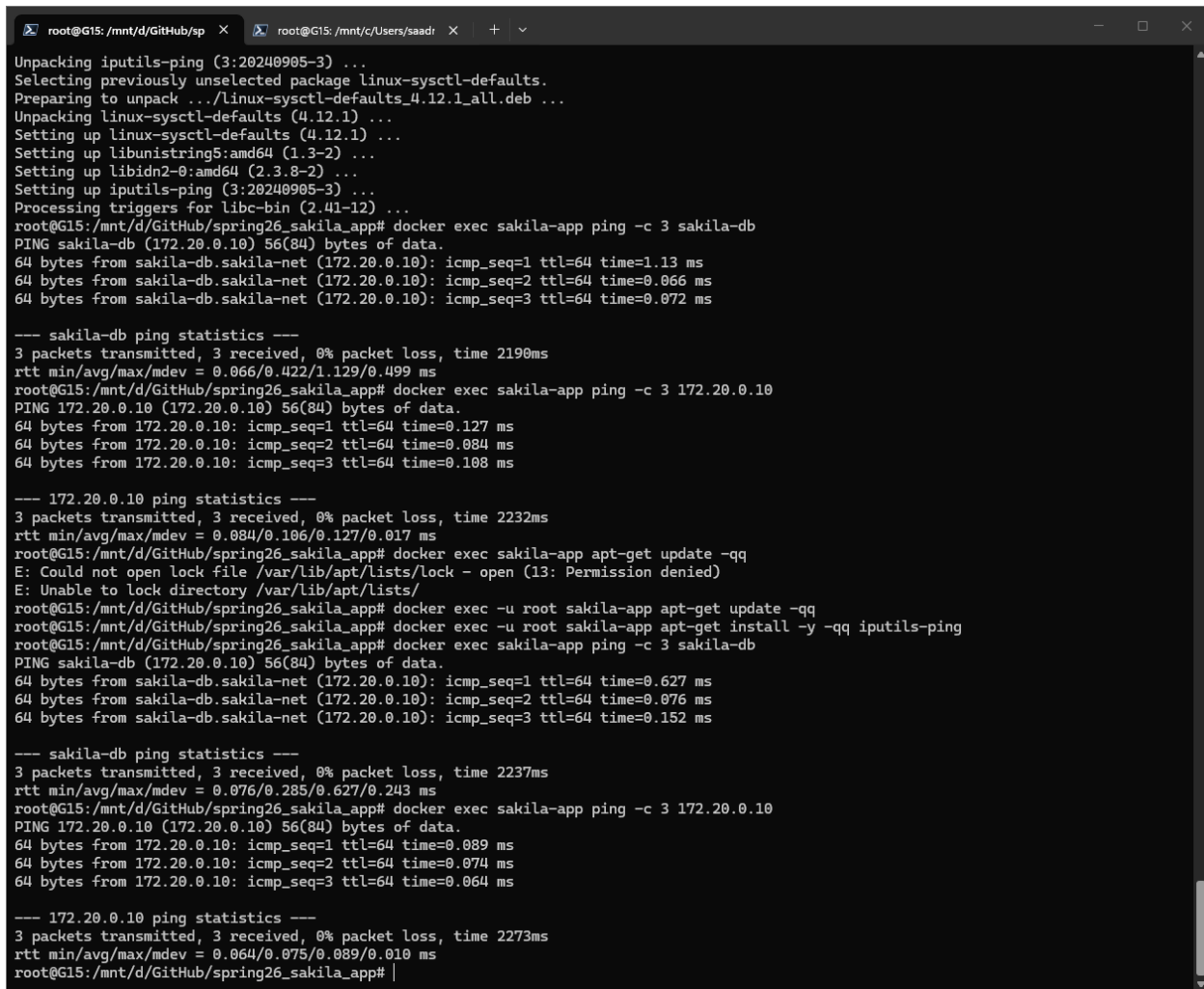
[Screenshot: B1 – docker inspect output showing Labels and Healthcheck sections]

B2 | Docker Networking

A custom bridge network named sakila-net was created with a defined subnet and gateway. The MySQL container was assigned a static IP (172.20.0.10) on this network. The Flask app container was connected to the same network and could reach the database by both container name and IP address.

To demonstrate network isolation, a separate network (sakila-isolated) was created. A test container on that network was unable to reach the sakila-db container,

confirming that containers on different networks cannot communicate unless explicitly connected.

A terminal window with two tabs. The first tab is titled 'root@G15: /mnt/d/GitHub/sp' and the second is 'root@G15: /mnt/c/Users/saad'. The terminal output shows the installation of 'linux-sysctl-defaults' and 'iputils-ping'. Then, a 'docker exec' command is used to run 'ping -c 3 sakila-db' from the 'sakila-app' container. The output shows three successful pings to 'sakila-db.sakila-net' (172.20.0.10) with times around 1.13ms, 0.066ms, and 0.072ms. Next, another 'docker exec' command runs 'ping -c 3 172.20.0.10' from the 'sakila-app' container, showing three successful pings to '172.20.0.10' with times around 0.127ms, 0.084ms, and 0.108ms. Finally, a third 'docker exec' command runs 'ping -c 3 172.20.0.10' from the 'sakila-app' container, showing three successful pings to '172.20.0.10' with times around 0.627ms, 0.076ms, and 0.152ms. The terminal also shows some error messages related to 'apt-get update' and 'apt-get install' when run from the 'sakila-app' container, indicating permission issues.

```
Unpacking iputils-ping (3:20240905-3) ...
Selecting previously unselected package linux-sysctl-defaults.
Preparing to unpack .../linux-sysctl-defaults_4.12.1_all.deb ...
Unpacking linux-sysctl-defaults (4.12.1) ...
Setting up linux-sysctl-defaults (4.12.1) ...
Setting up libunistring5:amd64 (1.3-2) ...
Setting up libidn2-0:amd64 (2.3.8-2) ...
Setting up iputils-ping (3:20240905-3) ...
Processing triggers for libc-bin (2.41-12) ...
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 sakila-db
PING sakila-db (172.20.0.10) 56(84) bytes of data.
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=1 ttl=64 time=1.13 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=2 ttl=64 time=0.066 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=3 ttl=64 time=0.072 ms

--- sakila-db ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2190ms
rtt min/avg/max/mdev = 0.066/0.422/1.129/0.499 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 172.20.0.10
PING 172.20.0.10 (172.20.0.10) 56(84) bytes of data.
64 bytes from 172.20.0.10: icmp_seq=1 ttl=64 time=0.127 ms
64 bytes from 172.20.0.10: icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from 172.20.0.10: icmp_seq=3 ttl=64 time=0.108 ms

--- 172.20.0.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2232ms
rtt min/avg/max/mdev = 0.084/0.106/0.127/0.017 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app apt-get update -qq
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -u root sakila-app apt-get update -qq
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -u root sakila-app apt-get install -y -qq iputils-ping
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 sakila-db
PING sakila-db (172.20.0.10) 56(84) bytes of data.
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=1 ttl=64 time=0.627 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=2 ttl=64 time=0.076 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=3 ttl=64 time=0.152 ms

--- sakila-db ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2237ms
rtt min/avg/max/mdev = 0.076/0.285/0.627/0.243 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 172.20.0.10
PING 172.20.0.10 (172.20.0.10) 56(84) bytes of data.
64 bytes from 172.20.0.10: icmp_seq=1 ttl=64 time=0.089 ms
64 bytes from 172.20.0.10: icmp_seq=2 ttl=64 time=0.074 ms
64 bytes from 172.20.0.10: icmp_seq=3 ttl=64 time=0.064 ms

--- 172.20.0.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2273ms
rtt min/avg/max/mdev = 0.064/0.075/0.089/0.010 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B2 - Successful ping from sakila-app to sakila-db (by name and IP)]

```
root@G15: /mnt/d/GitHub/sp x root@G15: /mnt/c/Users/saadri x + v
Setting up linux-sysctl-defaults (4.12.1) ...
Setting up libunistring5:amd64 (1.3-2) ...
Setting up libidn2-0:amd64 (2.3.8-2) ...
Setting up iputils-ping (3:20240905-3) ...
Processing triggers for libc-bin (2.41-12) ...
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 sakila-db
PING sakila-db (172.20.0.10) 56(84) bytes of data.
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=1 ttl=64 time=1.13 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=2 ttl=64 time=0.066 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=3 ttl=64 time=0.072 ms

--- sakila-db ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2190ms
rtt min/avg/max/mdev = 0.066/0.422/1.129/0.499 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 172.20.0.10
PING 172.20.0.10 (172.20.0.10) 56(84) bytes of data.
64 bytes from 172.20.0.10: icmp_seq=1 ttl=64 time=0.127 ms
64 bytes from 172.20.0.10: icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from 172.20.0.10: icmp_seq=3 ttl=64 time=0.108 ms

--- 172.20.0.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2232ms
rtt min/avg/max/mdev = 0.084/0.106/0.127/0.017 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app apt-get update -qq
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -u root sakila-app apt-get update -qq
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -u root sakila-app apt-get install -y -qq iputils-ping
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 sakila-db
PING sakila-db (172.20.0.10) 56(84) bytes of data.
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=1 ttl=64 time=0.627 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=2 ttl=64 time=0.076 ms
64 bytes from sakila-db.sakila-net (172.20.0.10): icmp_seq=3 ttl=64 time=0.152 ms

--- sakila-db ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2237ms
rtt min/avg/max/mdev = 0.076/0.285/0.627/0.243 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app ping -c 3 172.20.0.10
PING 172.20.0.10 (172.20.0.10) 56(84) bytes of data.
64 bytes from 172.20.0.10: icmp_seq=1 ttl=64 time=0.089 ms
64 bytes from 172.20.0.10: icmp_seq=2 ttl=64 time=0.074 ms
64 bytes from 172.20.0.10: icmp_seq=3 ttl=64 time=0.064 ms

--- 172.20.0.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2273ms
rtt min/avg/max/mdev = 0.064/0.075/0.089/0.010 ms
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app python -c "import pymysql"
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app python -c "import pymysql; c = pymysql.connect(host='sakila-db', user='root', password='admin', db='sakila'); print('MySQL CONNECTION SUCCESS'); c.close()"
MySQL CONNECTION SUCCESS
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B2 - MySQL Connection Test from sakila-app to sakila-db]

```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app python -c "import pymysql; c = pymysql.connect(host='sakila-db', user='root', password='admin', db='sakila'); print('MySQL CONNECTION SUCCESS'); c.close()"
MySQL CONNECTION SUCCESS
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker network inspect sakila-net
[
  {
    "Name": "sakila-net",
    "Id": "628ceefd13b89b62d7ccfd600b608aab20a4cb96e3fcd6edffdf272c7489cc9",
    "Created": "2026-04-27T07:41:46.732911194Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {},
    "Containers": {
      "380c33a79f553043e9107c53d69a86d4cd5d1a80324ee7b99e9336e5776a6149": {
        "Name": "sakila-db",
        "EndpointID": "6714b32463f1510fea35a45e4968b556d7f9c3764b566227812a858d69c0c274",
        "MacAddress": "a2:8f:2d:8d:a8:f0",
        "IPv4Address": "172.20.0.10/16",
        "IPv6Address": ""
      },
      "d2962855422a37c417cda8e628a78297ad419c9dd2bdc24dc1d2d0a89b8b8fd69": {
        "Name": "sakila-app",
        "EndpointID": "bd39de63f1ba5ecf640b0878af25190b240955ba30a95be4d91ca674248bf0ff",
        "MacAddress": "be:da:f6:bb:e5:17",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      }
    }
  }
],
```

[Screenshot: B2 - docker network inspect sakila-net showing both containers]

```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
},
"ConfigOnly": false,
"Options": {
  "com.docker.network.enable_ipv4": "true",
  "com.docker.network.enable_ipv6": "false"
},
"Labels": {},
"Containers": {
  "380c33a79f553043e9107c53d69a86d4cd5d1a80324ee7b99e9336e5776a6149": {
    "Name": "sakila-db",
    "EndpointID": "6714b32463f1510fea35a45e4968b556d7f9c3764b566227812a858d69c0c274",
    "MacAddress": "a2:8f:2d:8d:a8:f0",
    "IPv4Address": "172.20.0.10/16",
    "IPv6Address": ""
  },
  "d2962855422a37c417cda8e628a78297ad419c9dd2bdc24dc1d2d0a89b8b8fd69": {
    "Name": "sakila-app",
    "EndpointID": "bd39de63f1ba5ecf640b0878af25190b240955ba30a95be4d91ca674248bf0ff",
    "MacAddress": "be:da:f6:bb:e5:17",
    "IPv4Address": "172.20.0.2/16",
    "IPv6Address": ""
  }
}
}
"Status": {
  "IPAM": {
    "Subnets": {
      "172.20.0.0/16": {
        "IPsInUse": 5,
        "DynamicIPsAvailable": 65531
      }
    }
  }
}
}
]
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker network create sakila-isolated
Error response from daemon: network with name sakila-isolated already exists
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d --name test-isolated --network sakila-isolated alpine sleep 3600
docker: Error response from daemon: Conflict. The container name "/test-isolated" is already in use by container "fbad1dfa3a083ae6475a88c89d56fb2aeffb87bbdfb2a01c6c7f82d0dceble4". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec test-isolated ping -c 3 sakila-db
Error response from daemon: container fbad1dfa3a083ae6475a88c89d56fb2aeffb87bbdfb2a01c6c7f82d0dceble4 is not running
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker rm -f test-isolated
test-isolated
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d --name test-isolated --network sakila-isolated alpine sleep 3600
d8e4b0ea0134d3f3c2f5bc5df8f37067b4709500fa983f00966c90eaa7992922
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec test-isolated ping -c 3 sakila-db
ping: bad address 'sakila-db'
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B2 - Ping failure from test-isolated container proving network isolation]


```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
"380c33a79f553043e9107c53d69a86d4cd5d1a80324ee7b99e9336e5776a6149": {
  "Name": "sakila-db",
  "EndpointID": "6714b32463f1510fea35a45e4968b556d7f9c3764b566227812a858d69c0c274",
  "MacAddress": "a2:8f:2d:8d:a8:f0",
  "IPv4Address": "172.20.0.10/16",
  "IPv6Address": ""
},
"d2962855422a37c417cda8e628a78297ad419c9dd2bdc24dc1d2d0a89b8b8bf69": {
  "Name": "sakila-app",
  "EndpointID": "bd39de63f1ba5ecf640b0878af25190b240955ba30a95be4d91ca674248bf0ff",
  "MacAddress": "be:da:f6:bb:e5:17",
  "IPv4Address": "172.20.0.2/16",
  "IPv6Address": ""
}
},
"Status": {
  "IPAM": {
    "Subnets": {
      "172.20.0.0/16": {
        "IPsInUse": 5,
        "DynamicIPsAvailable": 65531
      }
    }
  }
}
}
]
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker network create sakila-isolated
Error response from daemon: network with name sakila-isolated already exists
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d --name test-isolated --network sakila-isolated alpine sleep 3600
docker: Error response from daemon: Conflict. The container name "/test-isolated" is already in use by container "fbad1dfa3a083ae6475a88c89d56fb2aeffb87bbdfb2a01c6c7f82d0dceble4". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec test-isolated ping -c 3 sakila-db
Error response from daemon: container fbad1dfa3a083ae6475a88c89d56fb2aeffb87bbdfb2a01c6c7f82d0dceble4 is not running
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker rm -f test-isolated
test-isolated
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d --name test-isolated --network sakila-isolated alpine sleep 3600
d8e4b8ea0134d3f3c2f5bc5df8f37067b4709500fa983f00966c90eaa7992922
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec test-isolated ping -c 3 sakila-db
ping: bad address 'sakila-db'
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-app cat /etc/hosts
127.0.0.1    localhost
::1         localhost ip6-localhost ip6-loopback
fe00::      ip6-localnet
ff00::      ip6-mcastprefix
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
172.20.0.2  d2962855422a
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B2 - /etc/hosts contents from inside the sakila-app container]

B3 | Docker Volumes and Data Persistence

A named Docker volume (sakila-mysql-data) was created and mounted to the MySQL container. A custom actor record with the student's name was inserted into the Sakila database. The container was then deleted and recreated using the same volume. The inserted record was still present after recreation, confirming that named volumes persist data outside the container lifecycle.

As a comparison, a second MySQL container was run without a named volume. After inserting a record and deleting the container, the record was lost when the

container was recreated, demonstrating why named volumes are necessary for persistent storage.

```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
local sakila-mysql-data
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker rm -f sakila-db
sakila-db
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d \
--name sakila-db \
--network sakila-net \
--ip 172.20.0.10 \
-v sakila-mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=admin \
-e MYSQL_DATABASE=sakila \
-e MYSQL_ROOT_HOST=% \
mysql:8.0
1d8e767c5fe8a31268d42ebbd6230a07d20d898a984d42432c74bf900053baa1
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker logs sakila-db 2>&1 | tail -5
2026-04-28T14:35:56.361950Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-04-28T14:35:56.362029Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-04-28T14:35:56.365723Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
2026-04-28T14:35:56.392101Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock
2026-04-28T14:35:56.392429Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker cp sakila-db/sakila-schema.sql sakila-db:/sakila-schema.sql
Successfully copied 24.3kB (transferred 26.1kB) to sakila-db:/sakila-schema.sql
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker cp sakila-db/sakila-data.sql sakila-db:/sakila-data.sql
Successfully copied 3.35MB (transferred 3.35MB) to sakila-db:/sakila-data.sql
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e "source /sakila-schema.sql;"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e "source /sakila-data.sql;"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-db mysql -uroot -padmin -e "SELECT COUNT(*) FROM sakila.actor;"
mysql: [Warning] Using a password on the command line interface can be insecure.
COUNT(*)
200
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e \
"USE sakila; INSERT INTO actor (first_name, last_name) VALUES ('Saad', 'Mughal');"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e "SELECT actor_id, first_name, last_name FROM sakila.actor ORDER BY actor_id DESC LIMIT 5;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+-----+-----+
| actor_id | first_name | last_name |
+-----+-----+-----+
| 201 | Saad | Mughal |
| 200 | THORA | TEMPLE |
| 199 | JULIA | FAWCETT |
| 198 | MARY | KEITEL |
| 197 | REESE | WEST |
+-----+-----+-----+
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B3 - Query result showing inserted actor record with actor_id before container deletion]

```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
COUNT(*)
200
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e \
"USE sakila; INSERT INTO actor (first_name, last_name) VALUES ('Saad', 'Mughal');"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e "SELECT actor_id, first_name, last_name FR
OM sakila.actor ORDER BY actor_id DESC LIMIT 5;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+-----+-----+
| actor_id | first_name | last_name |
+-----+-----+-----+
| 201 | Saad | Mughal |
| 200 | THORA | TEMPLE |
| 199 | JULIA | FAWCETT |
| 198 | MARY | KEITEL |
| 197 | REESE | WEST |
+-----+-----+-----+
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker rm -f sakila-db
sakila-db
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker ps -a | grep sakila-db
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d \
--name sakila-db \
--network sakila-net \
--ip 172.20.0.10 \
-v sakila-mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=admin \
mysql:8.0
4ffff57ad5fee9dbd6b86c78cb30e3c32cd95ab70fe3d4c96bddd5179f2397a38
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker logs sakila-db 2>&1 | tail -5
2026-04-28T14:38:50.617839Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-04-28T14:38:50.617921Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now
supported for this channel.
2026-04-28T14:38:50.621909Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the pa
th is accessible to all OS users. Consider choosing a different directory.
2026-04-28T14:38:50.645615Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var
/run/mysqld/mysqld.sock
2026-04-28T14:38:50.646073Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/ru
n/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-db mysql -uroot -padmin -e "SELECT actor_id, first_name, last_name FR
OM sakila.actor ORDER BY actor_id DESC LIMIT 5;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+-----+-----+
| actor_id | first_name | last_name |
+-----+-----+-----+
| 201 | Saad | Mughal |
| 200 | THORA | TEMPLE |
| 199 | JULIA | FAWCETT |
| 198 | MARY | KEITEL |
| 197 | REESE | WEST |
+-----+-----+-----+
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B3 – Same actor record still present after container was destroyed and recreated]

```
root@G15: /mnt/d/GitHub/sp x root@G15: /mnt/c/Users/saad/ x + v
-e MYSQL_DATABASE=sakila \
-e MYSQL_ROOT_HOST=% \
mysql:8.0
docker: Error response from daemon: Conflict. The container name "/sakila-no-vol" is already in use by container "960eb2ff868dad8a8fd70e1926e34e5c264419f79baf14947669be2b062728c". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker rm -f sakila-no-vol
sakila-no-vol
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d \
--name sakila-no-vol \
-e MYSQL_ROOT_PASSWORD=admin \
-e MYSQL_DATABASE=sakila \
-e MYSQL_ROOT_HOST=% \
mysql:8.0
2abc6b3f160d7b9fe0a51b780c391887821f6545546c55653464bc21cbee642f
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker ps -a | grep sakila-no-vol
2abc6b3f160d mysql:8.0 "docker-entrypoint.s..." 6 seconds ago Up 5 seconds 3306/tcp, 33060
/tcp
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker logs sakila-no-vol 2>&1 | tail -5
2026-04-28T14:40:24.750466Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-04-28T14:40:24.750544Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now
supported for this channel.
2026-04-28T14:40:24.753461Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the pa
th is accessible to all OS users. Consider choosing a different directory.
2026-04-28T14:40:24.771902Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var
/run/mysqld/mysqld.sock
2026-04-28T14:40:24.772126Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/ru
n/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker cp sakila-db/sakila-schema.sql sakila-no-vol:/sakila-schema.sql
Successfully copied 24.3kB (transferred 26.1kB) to sakila-no-vol:/sakila-schema.sql
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-no-vol mysql -uroot -padmin -e "source /sakila-schema.sql;"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-no-vol mysql -uroot -padmin -e "USE sakila; INSERT INTO actor (first_
name, last_name) VALUES ('NO', 'VOLUME');"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-no-vol mysql -uroot -padmin -e "SELECT actor_id, first_name, last_name FR
OM sakila.actor ORDER BY actor_id DESC LIMIT 3;"
mysql: [Warning] Using a password on the command line interface can be insecure.
actor_id first_name last_name
1 NO VOLUME
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker rm -f sakila-no-vol
sakila-no-vol
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker run -d --name sakila-no-vol -e MYSQL_ROOT_PASSWORD=admin -e MYSQL_DATABASE=sakila -e
MYSQL_ROOT_HOST=% mysql:8.0
3bf3d1d8e9fd50dfa06c342af63e0ceb7d97fc2e9df6d63d675aa3c71773f530
root@G15:/mnt/d/GitHub/spring26_sakila_app# sleep 30
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker exec sakila-no-vol mysql -uroot -padmin -e "SELECT COUNT(*) FROM sakila.actor;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1146 (42S02) at line 1: Table 'sakila.actor' doesn't exist
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B3 - Query showing data loss in the no-volume container after recreation]

```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saadri X + v
2026-04-28T14:40:24.750544Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-04-28T14:40:24.753461Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
2026-04-28T14:40:24.771902Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2026-04-28T14:40:24.772126Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysql.sock' port: 3306 MySQL Community Server - GPL.
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker cp sakila-db/sakila-schema.sql sakila-no-vol:/sakila-schema.sql
Successfully copied 24.3kB (transferred 26.1kB) to sakila-no-vol:/sakila-schema.sql
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-no-vol mysql -uroot -padmin -e "source /sakila-schema.sql;"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker exec -it sakila-no-vol mysql -uroot -padmin -e "USE sakila; INSERT INTO actor (first_name, last_name) VALUES ('NO', 'VOLUME');"
mysql: [Warning] Using a password on the command line interface can be insecure.
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker exec sakila-no-vol mysql -uroot -padmin -e "SELECT actor_id, first_name, last_name FROM sakila.actor ORDER BY actor_id DESC LIMIT 3;"
mysql: [Warning] Using a password on the command line interface can be insecure.
actor_id      first_name    last_name
1             NO            VOLUME
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker rm -f sakila-no-vol
sakila-no-vol
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker run -d --name sakila-no-vol -e MYSQL_ROOT_PASSWORD=admin -e MYSQL_DATABASE=sakila -e MYSQL_ROOT_HOST=% mysql:8.0
3bf3d1d8e9fd50dfa06c342af63e0ceb7d97fc2e9df6d63d675aa3c71773f539
root@G15: /mnt/d/GitHub/spring26_sakila_app# sleep 30
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker exec sakila-no-vol mysql -uroot -padmin -e "SELECT COUNT(*) FROM sakila.actor;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1146 (42S02) at line 1: Table 'sakila.actor' doesn't exist
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker volume inspect sakila-mysql-data
[
  {
    "CreatedAt": "2026-04-27T11:34:01Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/sakila-mysql-data/_data",
    "Name": "sakila-mysql-data",
    "Options": null,
    "Scope": "local"
  }
]
root@G15: /mnt/d/GitHub/spring26_sakila_app# docker volume ls
DRIVER      VOLUME NAME
local       4f0251f496fb7a607e00210be4e5561840b14ac9aade5a2898f0a416ad8ecee3
local       5a12aed396b44cd53a076a3bb2db9ecb48d47937036c483d3ff737acafd92434
local       08ccb5ff23d58bf56185c41e5be7e17aa4ef64088f4ae37f8df4d77cb0bd88d3
local       16e49ff8bb5849a106d6592d53e1a21cad1ca3e9785872ff15d611e248ea7020
local       080fef7359a1a212d9d8c8417fb8f60238e6cd6ea06792181eae3c3ccff07f1a
local       df440eed069aac221fa7f5ec219d26bfcfb39ae37bb3b92c31349b10ce54a9d3
local       sakila-mysql-data
root@G15: /mnt/d/GitHub/spring26_sakila_app#
```

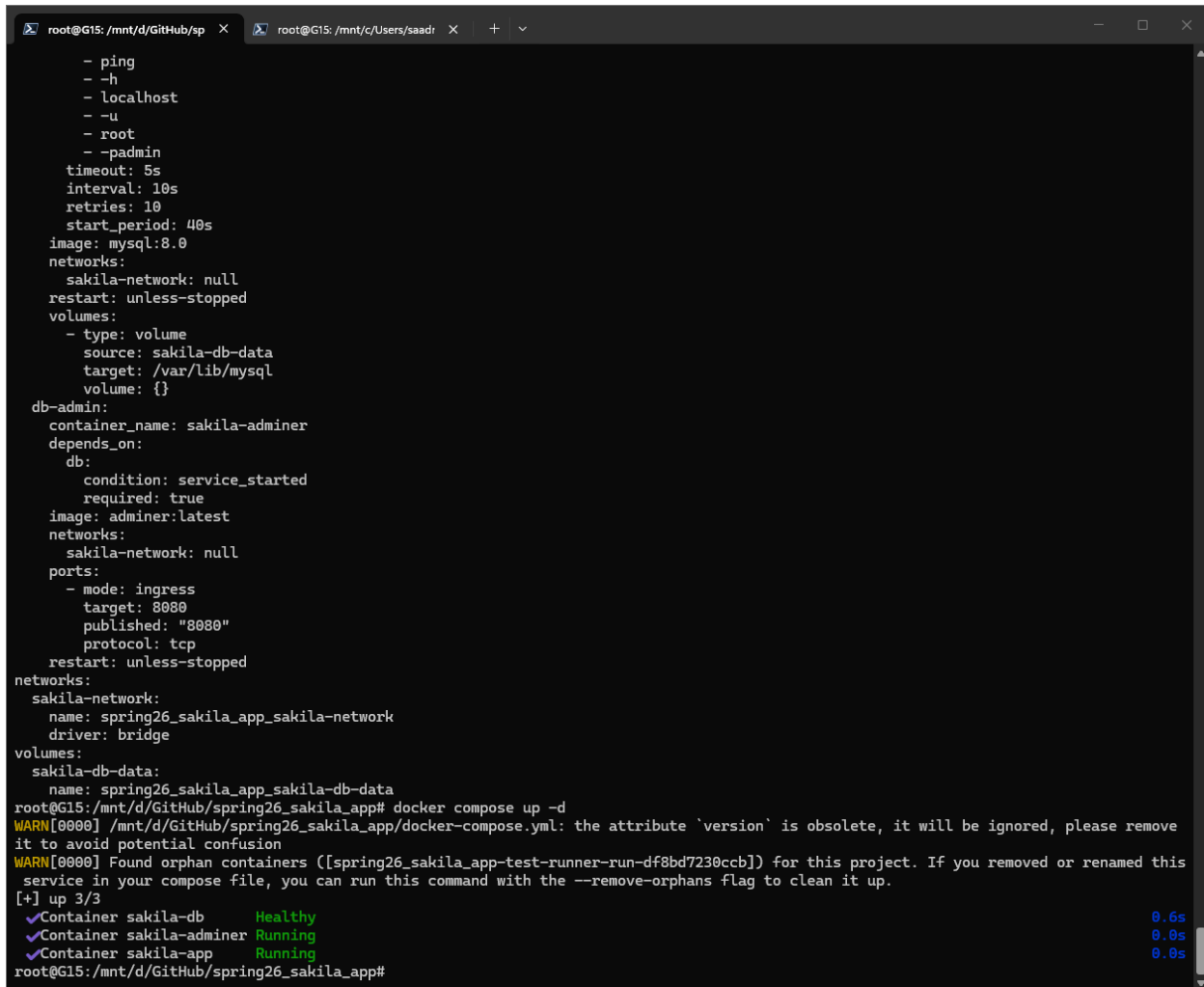
[Screenshot: B3 - docker volume inspect and docker volume ls output]

B4 | Docker Compose Orchestration

A docker-compose.yml file was created to orchestrate the full application stack. The file defined four services:

- db - MySQL database with a named volume and health check
- app - Flask application that depends on db being healthy
- db-admin - Adminer database UI accessible at port 8080
- test-runner - runs pytest and is only activated under the test profile

Environment variables were managed through a .env file, which was excluded from version control via .gitignore. A safe .env.example file with placeholder values was committed instead.

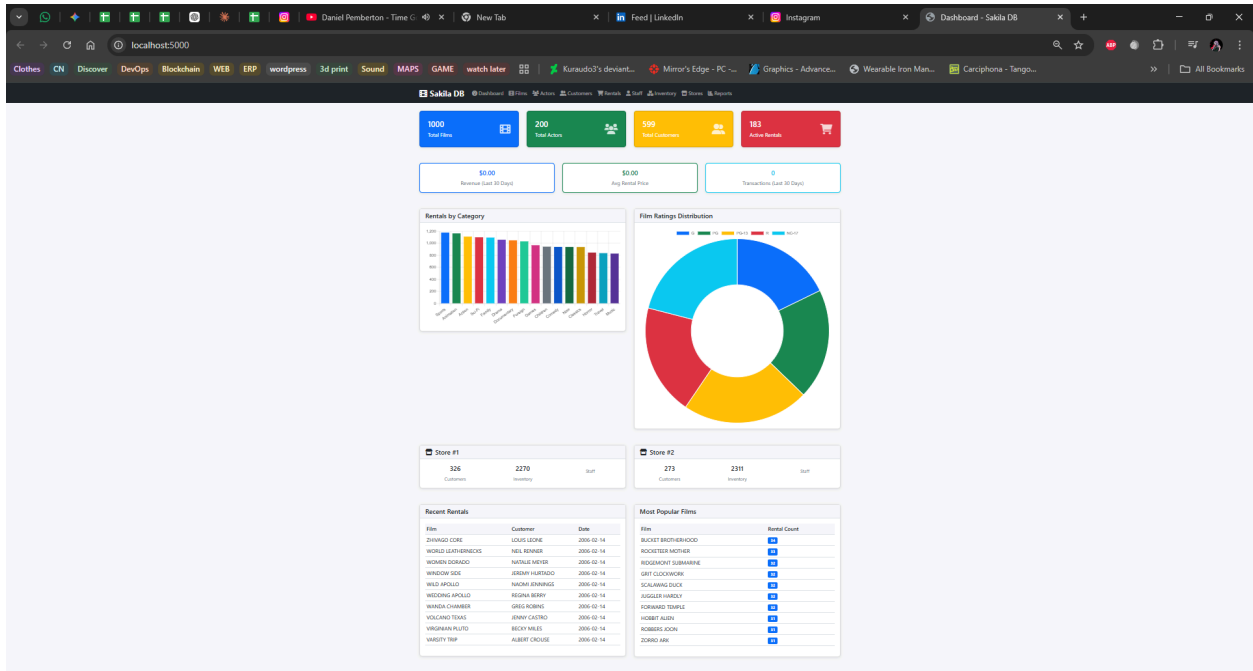


```
root@G15: /mnt/d/GitHub/sp x root@G15: /mnt/c/Users/saad x + v
- ping
- -h
- localhost
- -u
- root
- -padmin
timeout: 5s
interval: 10s
retries: 10
start_period: 40s
image: mysql:8.0
networks:
  sakila-network: null
restart: unless-stopped
volumes:
  - type: volume
    source: sakila-db-data
    target: /var/lib/mysql
    volume: {}
db-admin:
  container_name: sakila-adminer
  depends_on:
    db:
      condition: service_started
      required: true
  image: adminer:latest
  networks:
    sakila-network: null
  ports:
    - mode: ingress
      target: 8080
      published: "8080"
      protocol: tcp
  restart: unless-stopped
networks:
  sakila-network:
    name: spring26_sakila_app_sakila-network
    driver: bridge
volumes:
  sakila-db-data:
    name: spring26_sakila_app_sakila-db-data
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose up -d
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] Found orphan containers ([spring26_sakila_app-test-runner-run-df8bd7230ccb]) for this project. If you removed or renamed this service in your compose file, you can run this command with the --remove-orphans flag to clean it up.
[+] up 3/3
✓Container sakila-db      Healthy      0.6s
✓Container sakila-adminer Running      0.0s
✓Container sakila-app     Running      0.0s
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B4 - docker compose up -d showing all services created]

```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saad X + v
image: mysql:8.0
networks:
  sakila-network: null
restart: unless-stopped
volumes:
  - type: volume
    source: sakila-db-data
    target: /var/lib/mysql
    volume: {}
db-admin:
  container_name: sakila-adminer
  depends_on:
    db:
      condition: service_started
      required: true
  image: adminer:latest
  networks:
    sakila-network: null
  ports:
    - mode: ingress
      target: 8080
      published: "8080"
      protocol: tcp
  restart: unless-stopped
networks:
  sakila-network:
    name: spring26_sakila_app_sakila-network
    driver: bridge
volumes:
  sakila-db-data:
    name: spring26_sakila_app_sakila-db-data
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose up -d
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion
WARN[0000] Found orphan containers ([spring26_sakila_app-test-runner-run-df8bd7230ccb]) for this project. If you removed or renamed this
service in your compose file, you can run this command with the --remove-orphans flag to clean it up.
[+] up 3/3
✔Container sakila-db      Healthy      0.6s
✔Container sakila-adminer Running     0.0s
✔Container sakila-app     Running     0.0s
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose ps
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS                PORTS
sakila-adminer      adminer:latest      "entrypoint.sh docke..." db-admin    2 minutes ago  Up 2 minutes         0.0.0.0:8
080->8080/tcp, [::]:8080->8080/tcp
sakila-app          spring26_sakila_app-app "python app.py"        app         2 minutes ago  Up About a minute (unhealthy) 0.0.0.0:5
000->5000/tcp, [::]:5000->5000/tcp
sakila-db           mysql:8.0           "docker-entrypoint.s..." db          2 minutes ago  Up 2 minutes (healthy) 3306/tcp,
33060/tcp
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B4 - docker compose ps showing all services running/healthy]



[Screenshot: B4 - Browser showing Flask app at `http://localhost:5000`]

The screenshot shows the Adminer database management interface for the 'sakila' database. The interface includes a sidebar with navigation options and a main panel displaying the database schema.

Database: sakila

Tables and views

Table	Engine	Collation	Data Length	Index Length	Data Free	Auto Increment	Rows	Comment
actor	InnoDB	utf8mb4_0900_ai_ci	16,384	16,384	0	201	~ 200	
actor_info	View						7	
address	InnoDB	utf8mb4_0900_ai_ci	98,304	16,384	0	606	~ 603	
category	InnoDB	utf8mb4_0900_ai_ci	16,384	0	0	17	~ 16	
city	InnoDB	utf8mb4_0900_ai_ci	49,152	16,384	0	601	~ 600	
country	InnoDB	utf8mb4_0900_ai_ci	16,384	0	0	110	~ 109	
customer	InnoDB	utf8mb4_0900_ai_ci	81,920	49,152	0	600	~ 599	
customer_list	View						7	
film	InnoDB	utf8mb4_0900_ai_ci	196,608	81,920	0	1,001	~ 1,000	
film_actor	InnoDB	utf8mb4_0900_ai_ci	196,608	81,920	0	7	~ 5,462	
film_category	InnoDB	utf8mb4_0900_ai_ci	65,536	16,384	0	7	~ 1,000	
film_list	View						7	
film_text	InnoDB	utf8mb4_0900_ai_ci	180,224	16,384	0	7	~ 1,000	
inventory	InnoDB	utf8mb4_0900_ai_ci	180,224	196,608	0	4,582	~ 4,581	
language	InnoDB	utf8mb4_0900_ai_ci	16,384	0	0	7	~ 6	
nicer_but_slower_film_list	View						7	
payment	InnoDB	utf8mb4_0900_ai_ci	1,589,248	638,976	4,194,304	16,050	~ 15,813	
rental	InnoDB	utf8mb4_0900_ai_ci	1,589,248	1,196,032	4,194,304	16,050	~ 15,420	
sales_by_film_category	View						7	
sales_by_store	View						7	
staff	InnoDB	utf8mb4_0900_ai_ci	65,536	32,768	0	3	~ 2	
staff_list	View						7	

[Screenshot: B4 - Browser showing Adminer at `http://localhost:8080`]


```
root@G15: /mnt/d/GitHub/sp X root@G15: /mnt/c/Users/saadri X + v
sakila-adminer | [Tue Apr 28 14:53:31 2026] [::ffff:172.18.0.1]:57472 Accepted
sakila-adminer | [Tue Apr 28 14:53:31 2026] [::ffff:172.18.0.1]:57472 [403]: GET /?server=db&username=sakila_user
sakila-adminer | [Tue Apr 28 14:53:31 2026] [::ffff:172.18.0.1]:57472 Closing
sakila-adminer | [Tue Apr 28 14:53:52 2026] [::ffff:172.18.0.1]:46512 Accepted
sakila-adminer | [Tue Apr 28 14:53:52 2026] [::ffff:172.18.0.1]:46512 [302]: POST /?server=db&username=sakila_user
sakila-adminer | [Tue Apr 28 14:53:52 2026] [::ffff:172.18.0.1]:46512 Closing
sakila-adminer | [Tue Apr 28 14:53:52 2026] [::ffff:172.18.0.1]:46522 Accepted
sakila-adminer | [Tue Apr 28 14:53:52 2026] [::ffff:172.18.0.1]:46522 [403]: GET /?username=user&db=sakila
sakila-adminer | [Tue Apr 28 14:53:52 2026] [::ffff:172.18.0.1]:46522 Closing
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53354 Accepted
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53354 [302]: POST /?username=user&db=sakila
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53354 Closing
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53362 Accepted
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53362 [200]: GET /?server=db&username=root
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53362 Closing
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53372 Accepted
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53372 [200]: GET /?file=jush.js&version=5.4.2
sakila-adminer | [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53372 Closing
sakila-adminer | [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53374 Accepted
sakila-adminer | [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53374 [200]: GET /?server=db&username=root&db=sakila
sakila-adminer | [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53374 Closing
sakila-adminer | [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53376 Accepted
sakila-adminer | [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53376 [200]: GET /?server=db&username=root&db=sakila&script=db
sakila-adminer | [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53376 Closing
^C
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose --profile test run test-runner
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion
WARN[0000] Found orphan containers ([spring26_sakila_app-test-runner-run-df8bd7230ccb]) for this project. If you removed or renamed this
service in your compose file, you can run this command with the --remove-orphans flag to clean it up.
[+] 1/1t 1/11
✓Container sakila-db Running 0.0s
Container sakila-db Waiting
Container sakila-db Healthy
Container spring26_sakila_app-test-runner-run-7ad626419c54 Creating
Container spring26_sakila_app-test-runner-run-7ad626419c54 Created
===== test session starts =====
platform linux -- Python 3.9.25, pytest-8.3.3, pluggy-1.6.0 -- /usr/local/bin/python
cachedir: .pytest_cache
rootdir: /app
collected 6 items

tests/test_api.py::test_actor_api_returns_json_error_when_db_fails PASSED [ 16%]
tests/test_api.py::test_film_api_returns_json_error_when_db_fails PASSED [ 33%]
tests/test_basic.py::test_app_exists PASSED [ 50%]
tests/test_basic.py::test_app_name PASSED [ 66%]
tests/test_basic.py::test_config_defaults_exist PASSED [ 83%]
tests/test_basic.py::test_dashboard_route_handles_database_error PASSED [100%]

===== 6 passed in 0.38s =====
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B4 - test-runner container executing pytest]

```
root@G15: /mnt/d/GitHub/sp x root@G15: /mnt/c/Users/saadri x + v
sakila-adminer [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53362 [200]: GET /?server=db&username=root
sakila-adminer [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53362 Closing
sakila-adminer [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53372 Accepted
sakila-adminer [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53372 [200]: GET /?file=jush.js&version=5.4.2
sakila-adminer [Tue Apr 28 14:54:16 2026] [::ffff:172.18.0.1]:53372 Closing
sakila-adminer [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53374 Accepted
sakila-adminer [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53374 [200]: GET /?server=db&username=root&db=sakila
sakila-adminer [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53374 Closing
sakila-adminer [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53376 Accepted
sakila-adminer [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53376 [200]: GET /?server=db&username=root&db=sakila&script=db
sakila-adminer [Tue Apr 28 14:54:22 2026] [::ffff:172.18.0.1]:53376 Closing
^C
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose --profile test run test-runner
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion
WARN[0000] Found orphan containers ([spring26_sakila_app-test-runner-run-df8bd7230ccb]) for this project. If you removed or renamed this
service in your compose file, you can run this command with the --remove-orphans flag to clean it up.
[+] 1/1t 1/11
✓Container sakila-db Running 0.0s
Container sakila-db Waiting
Container sakila-db Healthy
Container spring26_sakila_app-test-runner-run-7ad626419c54 Creating
Container spring26_sakila_app-test-runner-run-7ad626419c54 Created
===== test session starts =====
platform linux -- Python 3.9.25, pytest-8.3.3, pluggy-1.6.0 -- /usr/local/bin/python
cachedir: .pytest_cache
rootdir: /app
collected 6 items

tests/test_api.py::test_actor_api_returns_json_error_when_db_fails PASSED [ 16%]
tests/test_api.py::test_film_api_returns_json_error_when_db_fails PASSED [ 33%]
tests/test_basic.py::test_app_exists PASSED [ 50%]
tests/test_basic.py::test_app_name PASSED [ 66%]
tests/test_basic.py::test_config_defaults_exist PASSED [ 83%]
tests/test_basic.py::test_dashboard_route_handles_database_error PASSED [100%]

===== 6 passed in 0.38s =====
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose down
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion
[+] down 4/4
✓Container sakila-adminer Removed 0.4s
✓Container sakila-app Removed 0.5s
✓Container sakila-db Removed 1.4s
✓Network spring26_sakila_app_sakila-network Removed 0.4s
root@G15:/mnt/d/GitHub/spring26_sakila_app# docker compose down -v
WARN[0000] /mnt/d/GitHub/spring26_sakila_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion
[+] down 1/1
✓Volume spring26_sakila_app_sakila-db-data Removed 0.0s
root@G15:/mnt/d/GitHub/spring26_sakila_app#
```

[Screenshot: B4 - docker compose down -v output showing containers and volumes removed]

Part C: GitHub Actions CI/CD

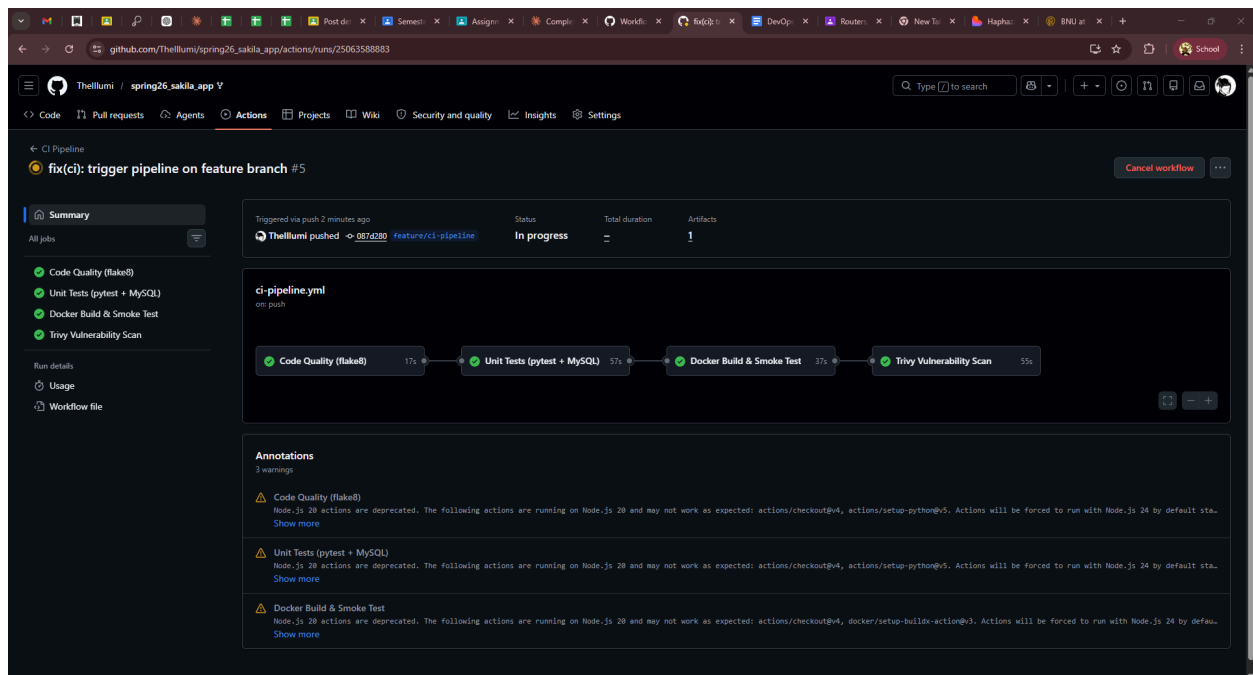
Part C implemented automated CI/CD pipelines using GitHub Actions. A multi-stage CI pipeline was created along with a separate deployment workflow. The pipelines were triggered on pull requests and branch pushes.

C1 | Multi-Stage CI Pipeline

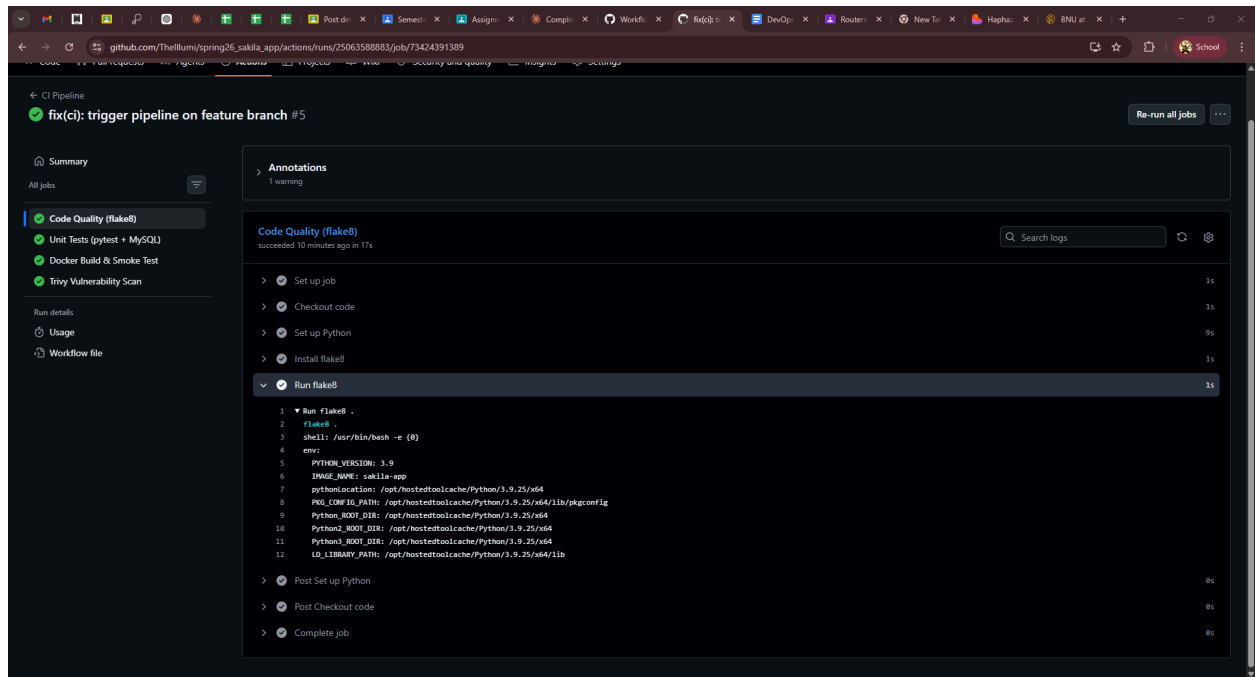
The CI pipeline (ci-pipeline.yml) was created with four sequential jobs:

- Lint - runs flake8 to check code style; configured as non-blocking for legacy code
- Test - runs pytest against a MySQL service container with the Sakila database imported
- Build - builds the Docker image and runs a smoke test
- Security Scan - scans the image using Trivy and uploads the report as a build artifact

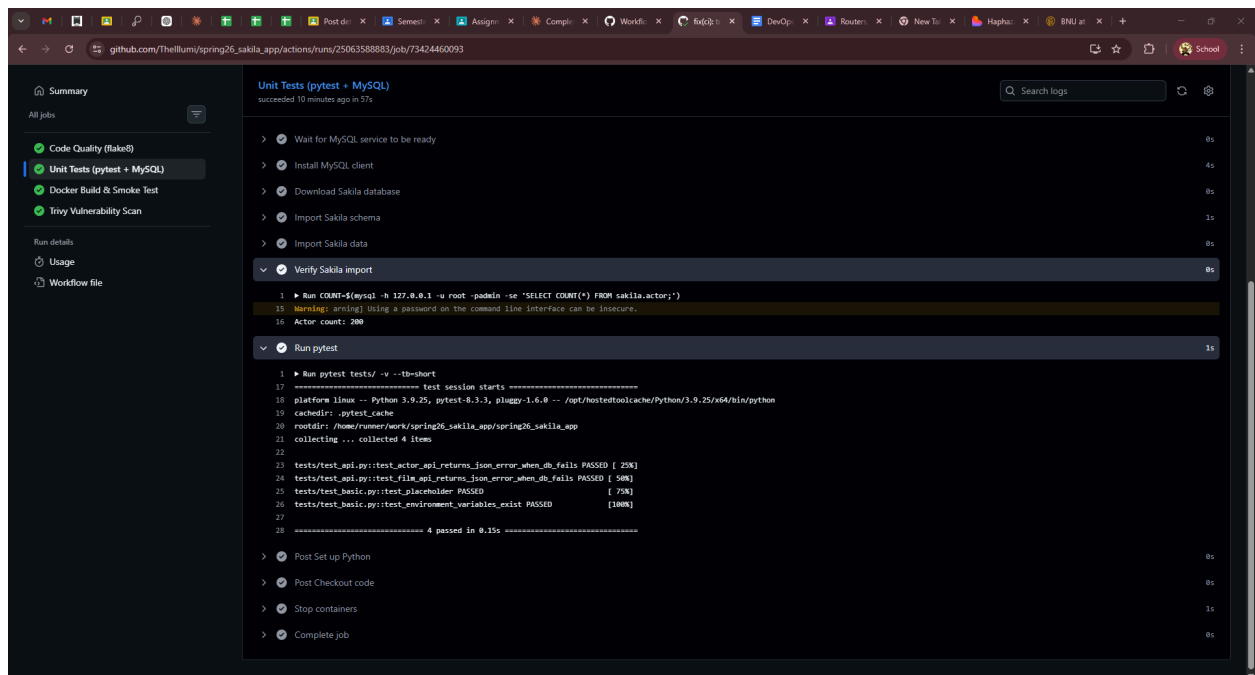
During initial runs, the lint job failed due to formatting issues in the legacy app.py file. This was fixed by making the lint step non-blocking. The test job also failed initially because Python could not find the app module; this was resolved by setting the PYTHONPATH environment variable in the workflow.



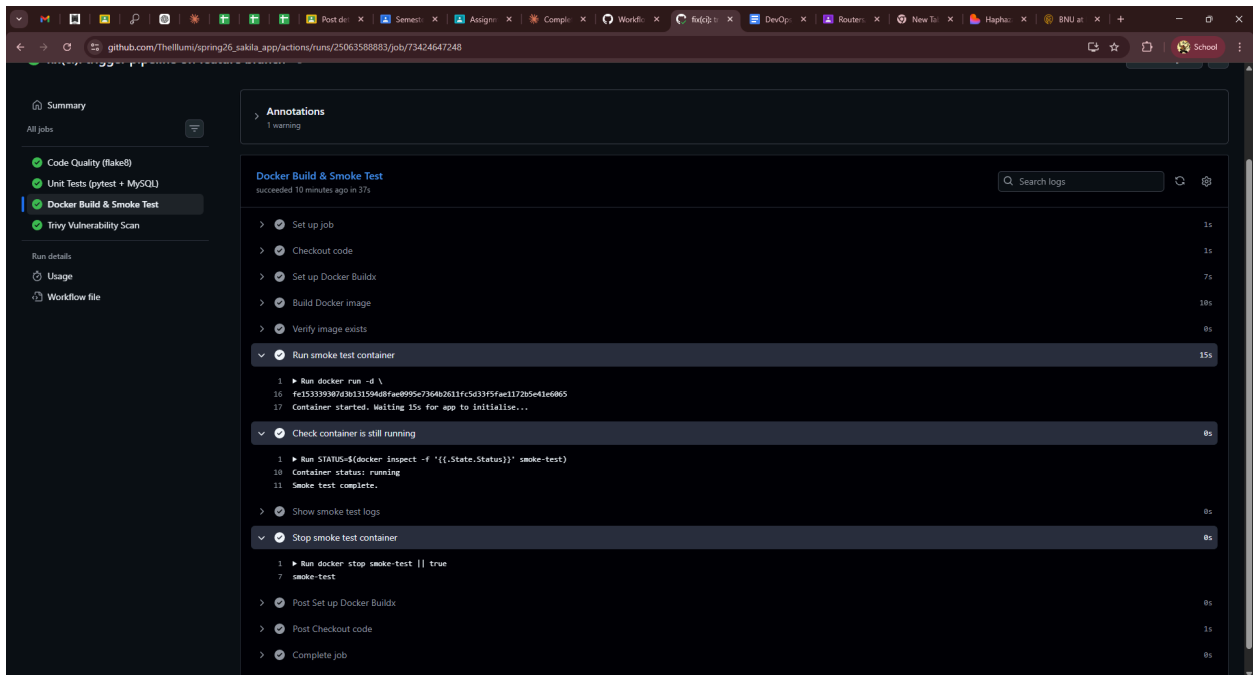
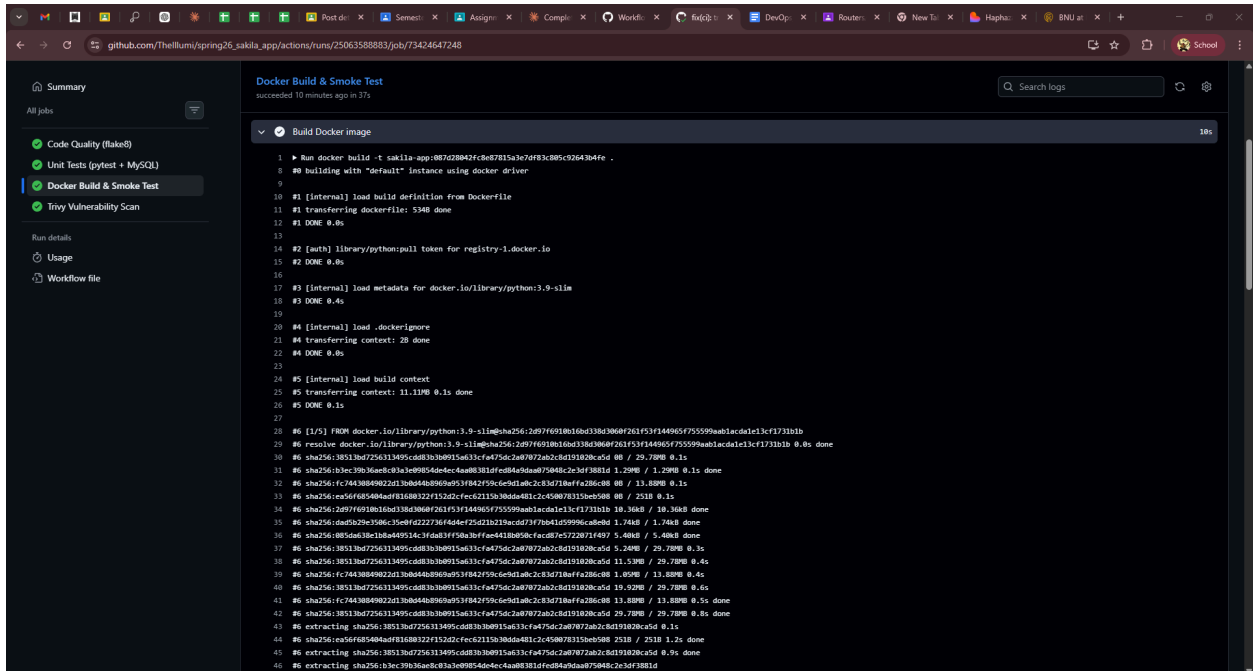
[Screenshot: C1 - GitHub Actions page showing all 4 jobs (lint, test, build, security-scan) passing]



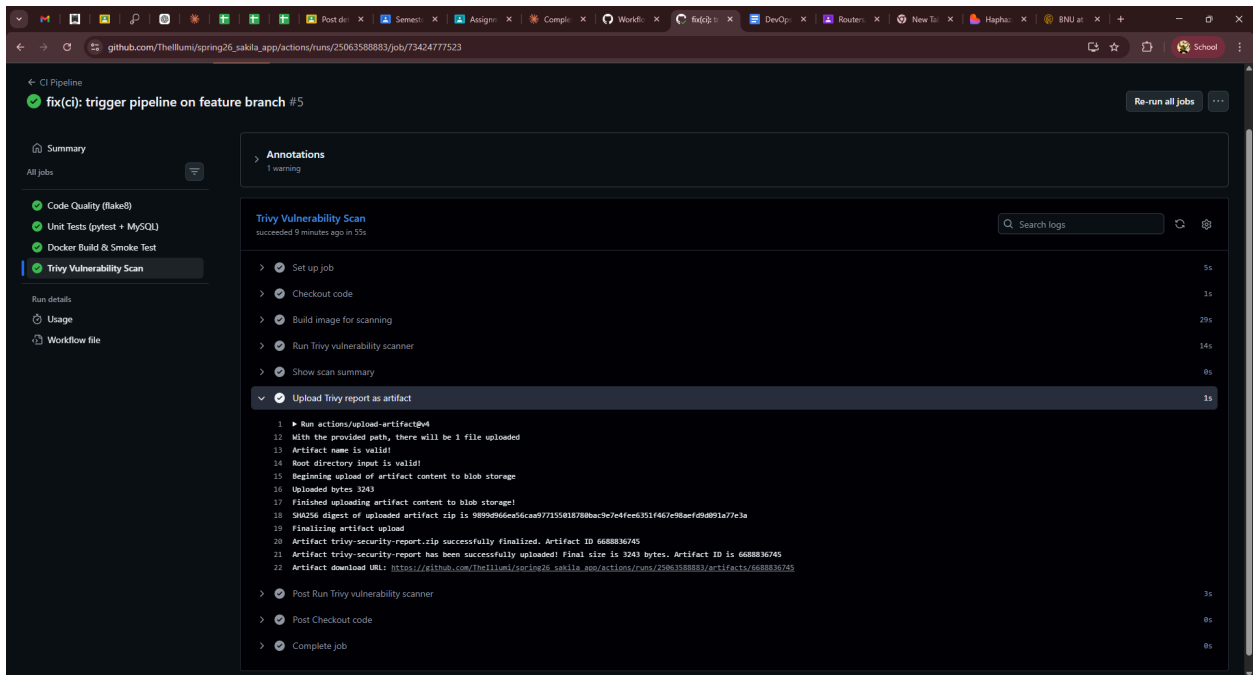
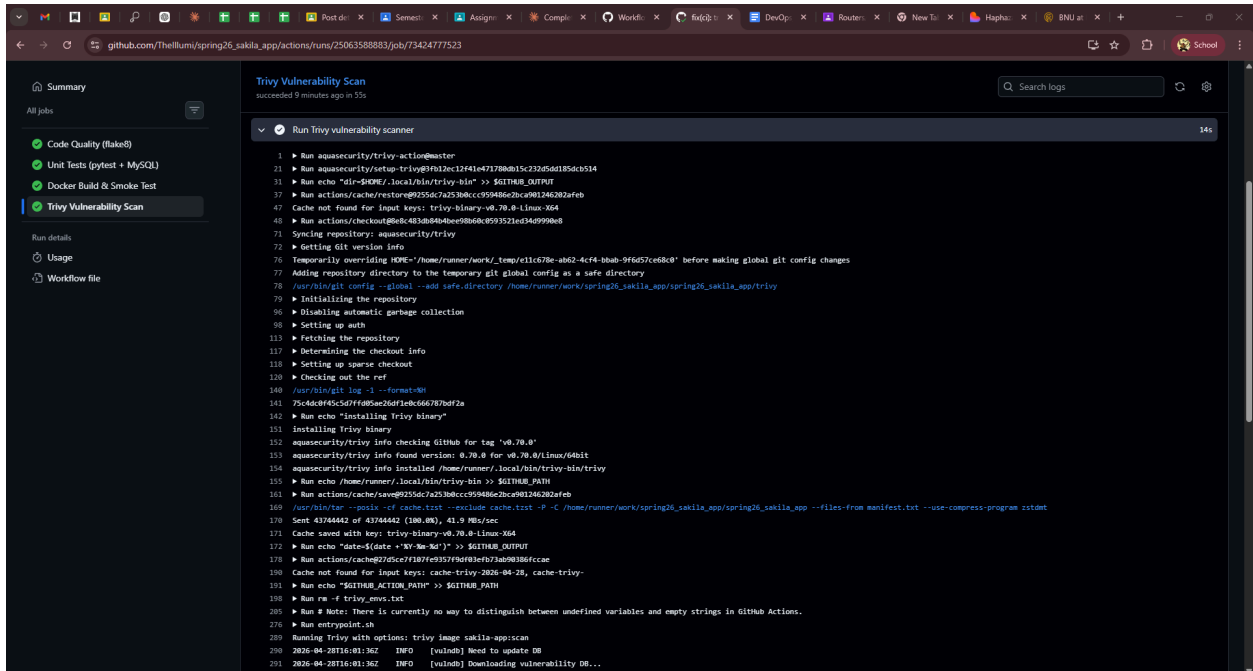
[Screenshot: C1 — Lint job logs showing flake8 output]



[Screenshot: C1 — Test job logs showing pytest results]



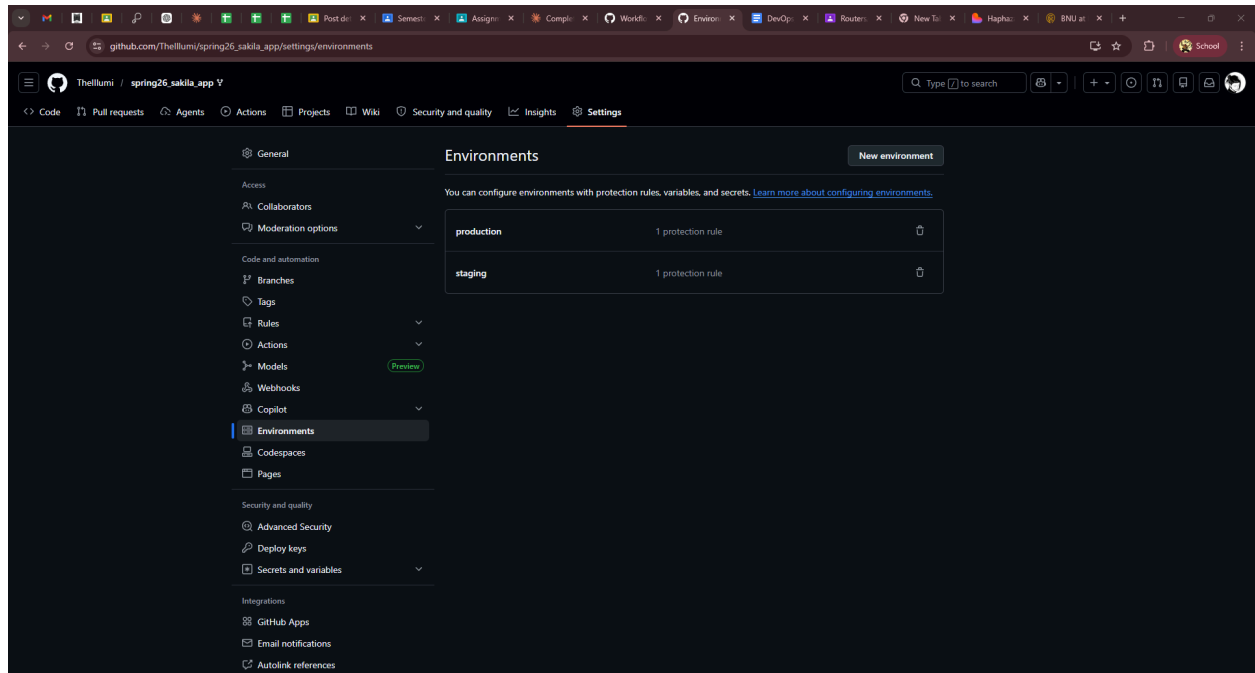
[Screenshot: C1 — Build job logs showing Docker image built and smoke test run]



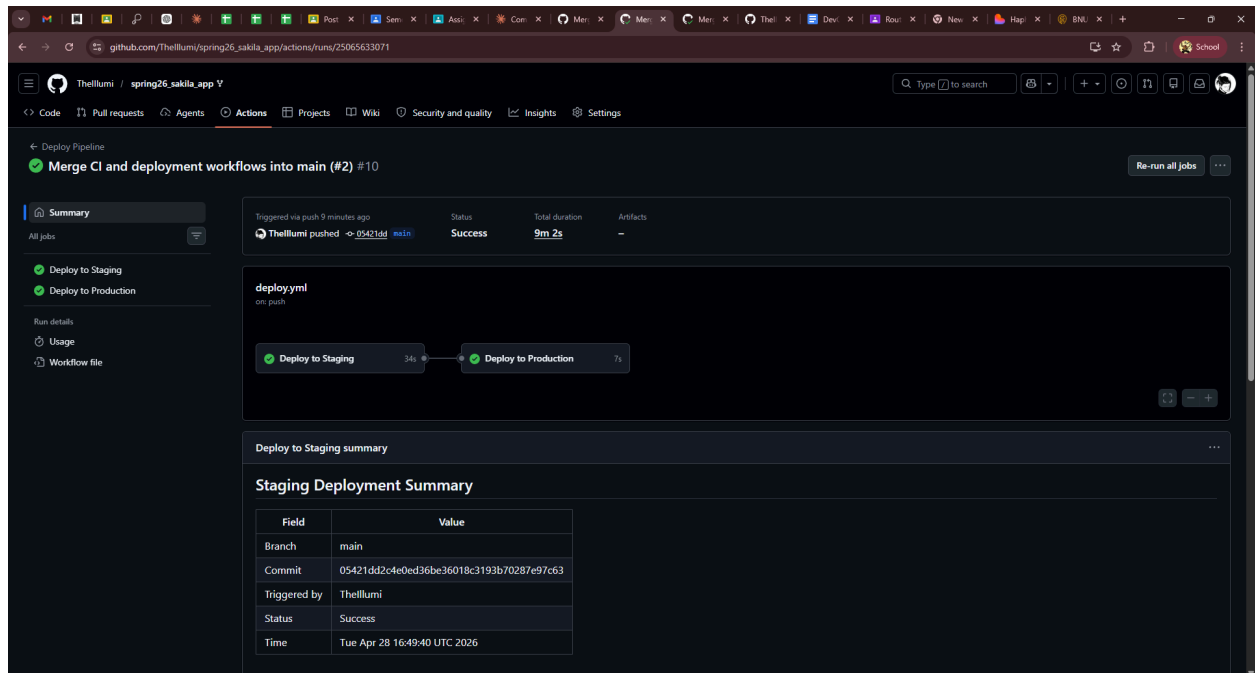
[Screenshot: C1 — Security scan job showing Trivy output and uploaded artifact]

C2 | Deploy Pipeline

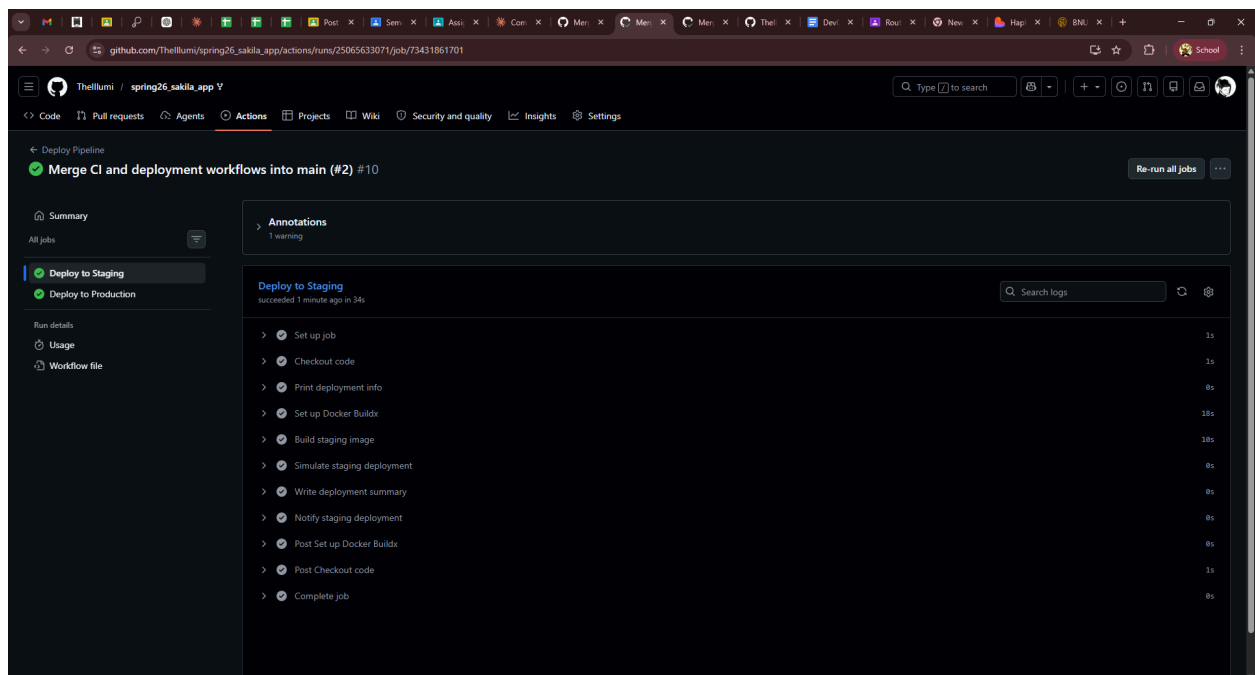
A separate deploy workflow (deploy.yml) was added to simulate staging and production deployment. The staging job runs on all pushes to the feature branch and prints deployment confirmation messages. The production job is gated to only run when a version tag (e.g. v2.0.0) is pushed, representing a controlled release process.

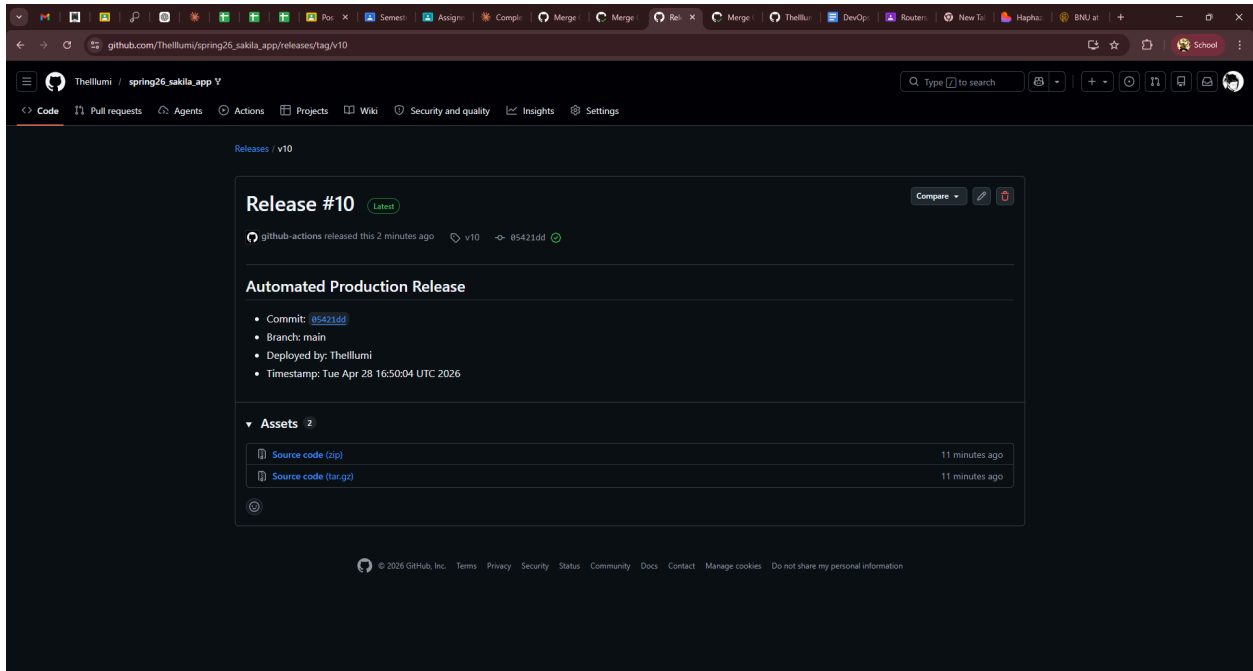


[Screenshot: C2 - Deploy Pipeline workflow runs listed in GitHub Actions]



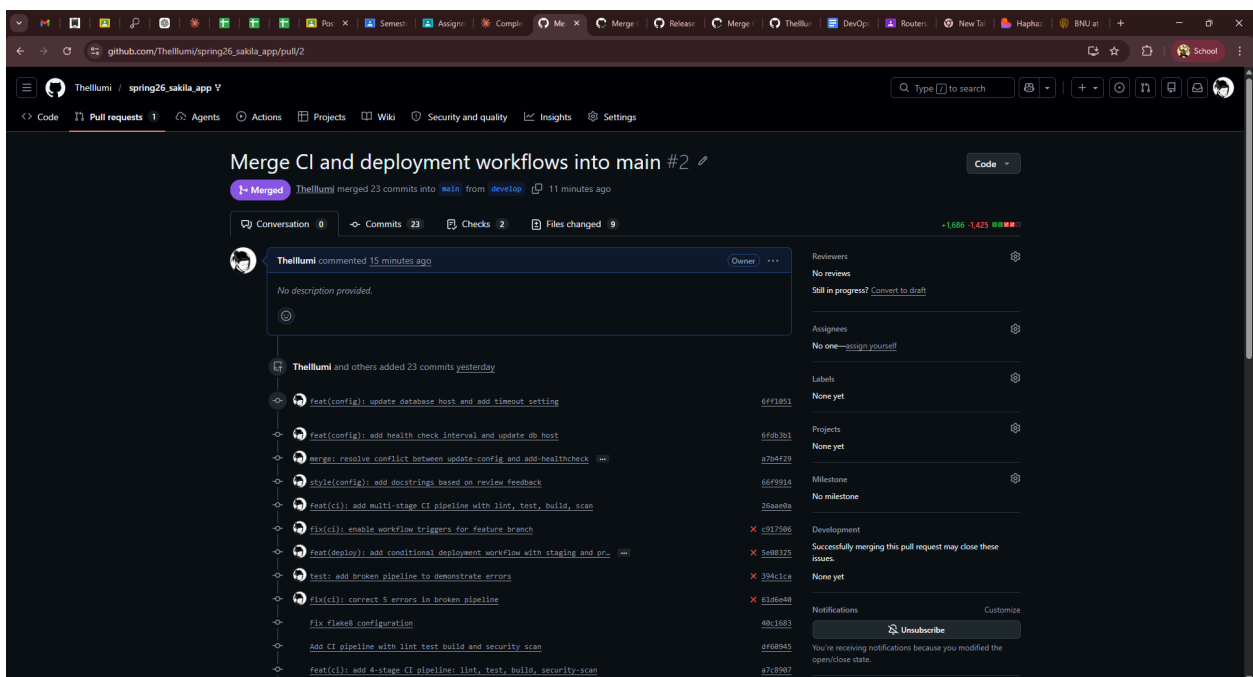
[Screenshot: C2 – Staging deployment job logs showing successful deployment simulation]

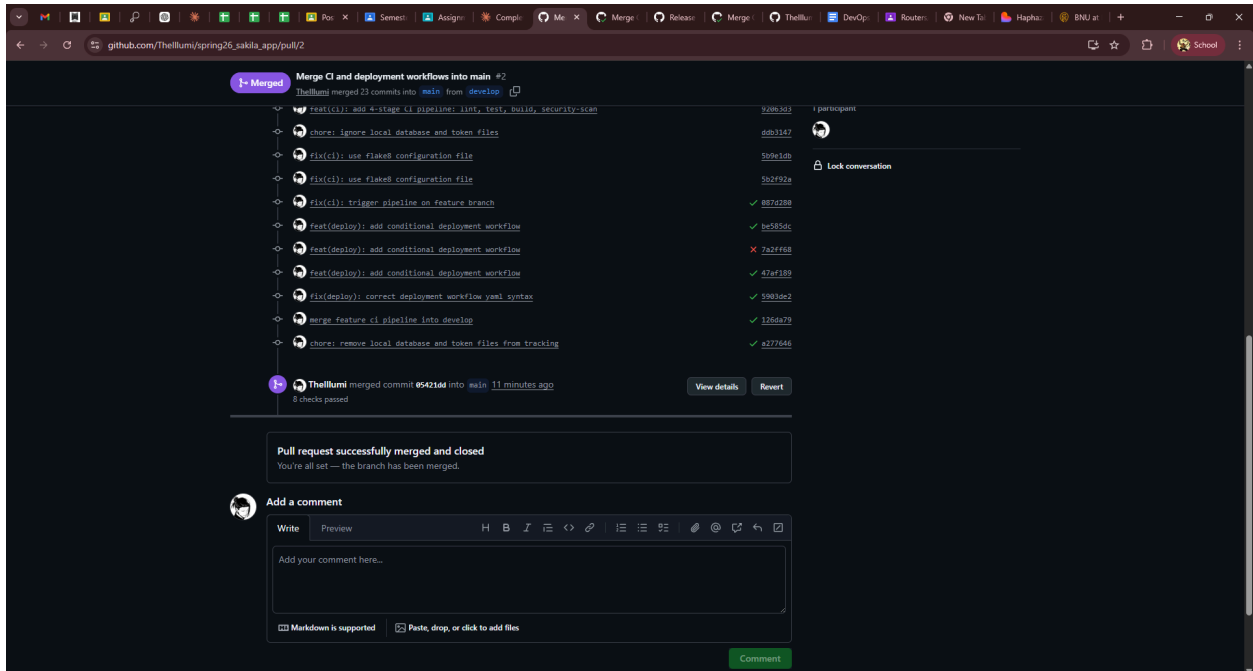




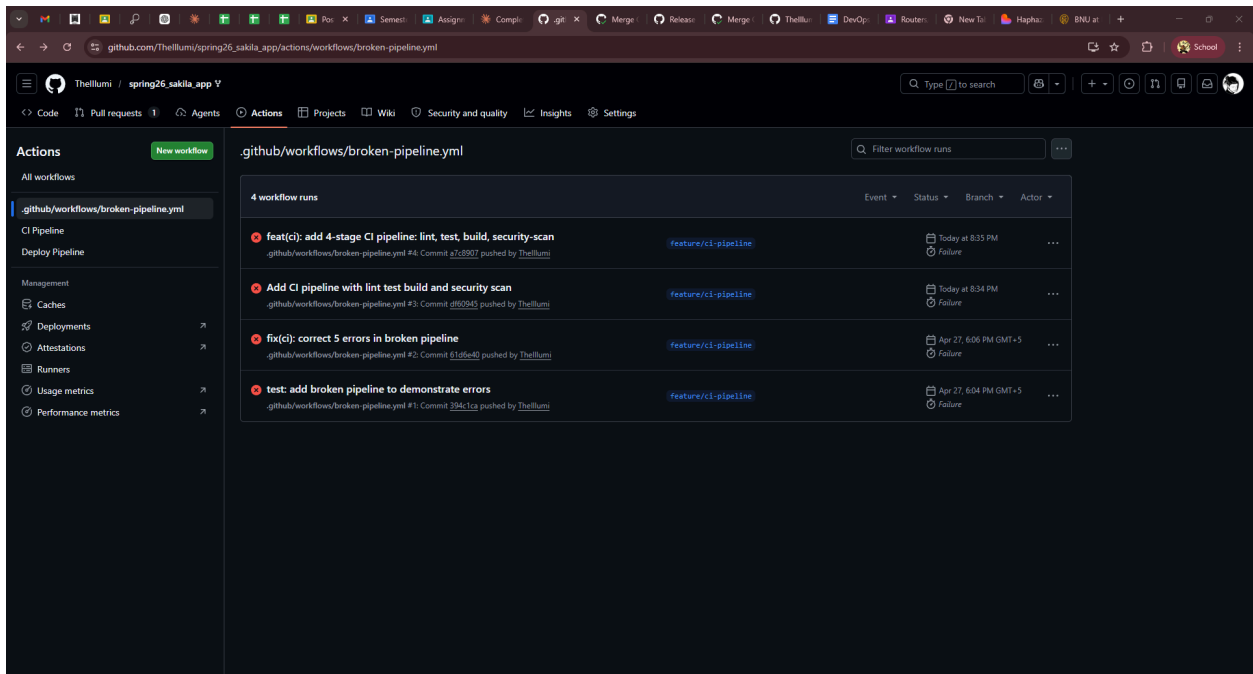
C3 | Part C Merge and Tagging

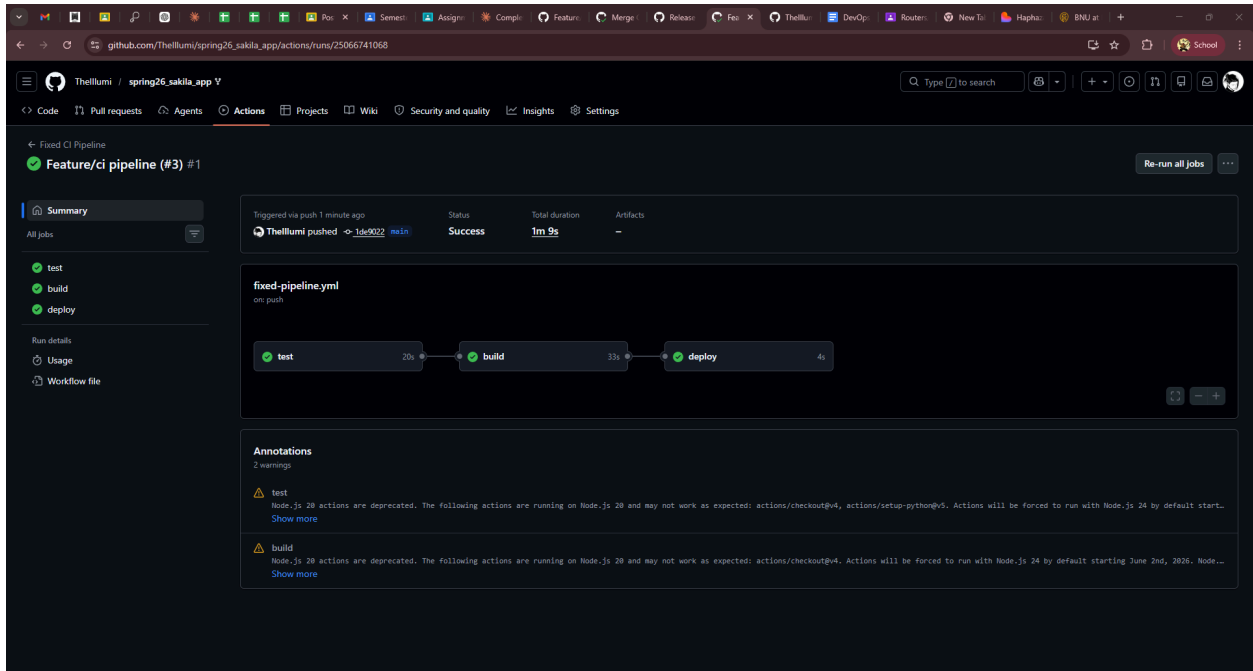
After the CI/CD workflows were verified, the feature/ci-pipeline branch was merged into develop via a pull request. A version tag v2.0.0 was created and pushed to mark the completion of Part C.





[Screenshot: C3 – Part C pull request merged into develop]





In the debugging challenge, I created an intentionally broken GitHub Actions workflow containing five common CI/CD configuration errors. The workflow failed as expected, confirming that the issues disrupted pipeline execution.

The first error was an invalid runner name: `runs-on: ubuntu`. GitHub-hosted runners require valid labels such as `ubuntu-latest`, so this was corrected accordingly.

The second error was the Python version written as `3.9` without quotes. YAML may interpret this as a number, so it was corrected to `'3.9'` to ensure `setup-python` reads it as a version string.

The third error was that the build job did not depend on the test job, allowing builds to run even if tests failed. This was fixed by adding `needs: test`.

The fourth error was that the deploy job depended on test instead of build. Deployment should only occur after a successful build, so it was changed to `needs: build`.

The fifth error was the branch trigger written as `branches: main`. GitHub Actions expects branch filters as a list, so it was corrected to `branches: [main]`.

After applying these fixes, the pipeline executed successfully, with the test, build, and deploy jobs completing in the correct sequence.

Challenges Faced and Fixes

Part A

A merge conflict occurred in `config.py` because two feature branches modified `MYSQL_HOST` with different values. The conflict was resolved manually by reviewing the conflict markers, selecting the correct host value, and combining both new configuration variables into the final file.

Part B

The MySQL container was sometimes not ready when dependent commands executed. This was resolved by waiting for the container to initialize and verifying readiness through docker logs. On Windows, Git Bash converted `/etc/hosts` into a Windows path, which was worked around by using `//etc/hosts`.

Part C

The CI lint job initially failed because flake8 reported many style issues in legacy code. To prevent blocking the pipeline, the lint step was configured as non-blocking. The test job failed separately because Python could not locate the app module until `PYTHONPATH` was explicitly set in the workflow environment.

Conclusion

This assignment demonstrated a complete DevOps workflow covering version control, containerization, and CI/CD automation. Key outcomes included:

- A structured Git branching strategy with protected branches and conflict resolution
- An optimized Docker image with networking and persistent storage

- Full-stack orchestration using Docker Compose
- An automated pipeline for linting, testing, Docker builds, security scanning, and deployment simulation

The project showed how each layer of a DevOps pipeline connects, from a developer's first commit to a reproducible, tested, and deployable container image.